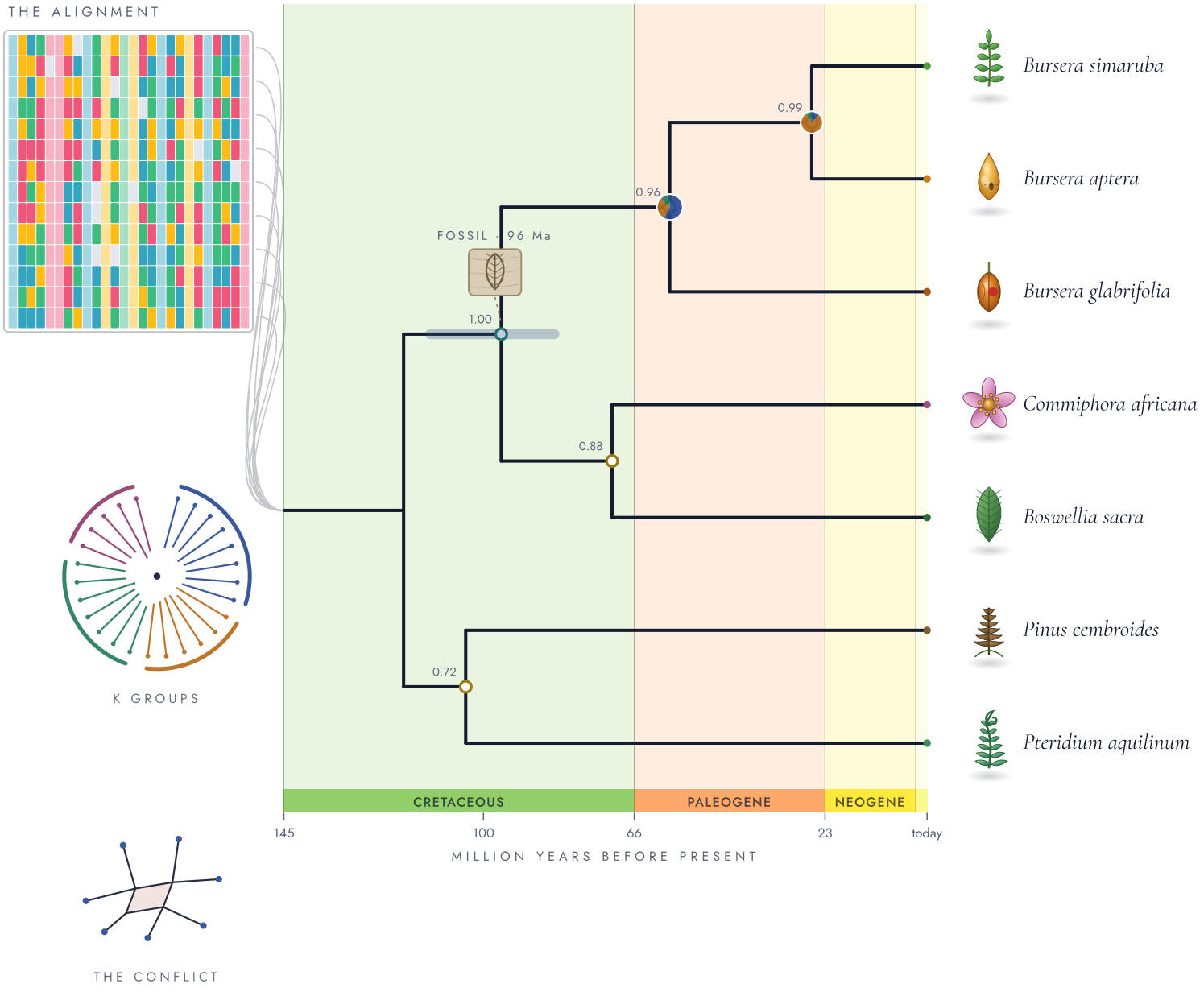
PhylogenyPro

User's Manual

Manual de usuario

FROM THE ALIGNMENT TO THE DATED TREE



Luis Ángel Barrera-Guzmán

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PhylogenyPro

User's Manual · Molecular phylogenetics, from the alignment to the dated tree

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PROGRAM	PhylogenyPro, version 1.4.0 (September 2026)
MANUAL	First English edition, 2026, translated from the first Spanish edition. Describes version 1.4.0 of the program.

PROGRAM LICENSE	GNU General Public License, version 3 or later (GPL-3.0-or-later). The full text comes in the LICENSE file.
MANUAL RIGHTS	© 2026 Luis Ángel Barrera-Guzmán. All rights reserved.

INTERFACE LANGUAGE	The program is in **Spanish and English**, and you switch from one to the other with the ES and EN buttons in the upper right corner, without losing your work. In this manual buttons and menus are written as they appear on screen in English, for example Build the tree.

HOW TO CITE THE PROGRAM

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The program is deposited in Zenodo under the concept DOI 10.5281/zenodo.23005815, which always resolves to the latest version. Machine-readable citation data are in CITATION.cff and codemeta.json, at the root of the program.

> **PRIVACY**
>
> PhylogenyPro runs entirely inside your browser. The files you open are read on your computer and **are never sent to any server**. You can analyze unpublished sequences, data under editorial embargo or material subject to an access agreement without any risk of them leaving your machine. The photographs you add to the tips of a tree are also stored in your browser and do not travel anywhere either.

The illustrations, diagrams and screenshots in this manual are original and were generated with the program itself or with code written for the manual. The example data sets are described in Section I.7: the sequences are **simulated** on a known tree, so that each example carries its own correct answer; the species names are real, the sequences are not. The provenance of the scientific constants built into the program—the amino-acid substitution matrices, the BLOSUM62 matrix, the genetic code and the geological time scale—is documented in the PROCEDENCIA.md file.

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INTRODUCTION · BEFORE THE BLOCKS

A tree is a hypothesis

What the program is, which questions it answers, what data it accepts, how to open it and how to work through it. It includes the six ideas you need to understand the rest of the manual, the description of the example data sets—simulated on purpose, so that they carry their own answer—and a frank list of what the program does not do.

I.1What is PhylogenyPro?

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I.1 What is PhylogenyPro?

PhylogenyPro is an application that carries out a complete phylogenetic study inside the browser. It opens with a double click on a file, it is not installed, it asks for no account or license, and it does not send your data anywhere. On the other hand, it is not a viewer: it really computes. It aligns the sequences, chooses the substitution model, searches for the tree by parsimony, by distances, by maximum likelihood and by Bayesian inference, measures the support of each branch, tests the molecular clock, dates the nodes with fossils, fits diversification models, reconstructs ancestral characters and areas, compares trees that disagree and, at the end, writes the methods section with the numbers from your run and assembles a package with everything so that someone else can repeat it.

It is meant for anyone who needs to do a phylogenetic analysis properly and does not want—or cannot—set up a tower of command-line programs to get there. A graduate student with a first gene, a taxonomist who wants to date a radiation, a professor who has to teach maximum likelihood on Tuesday.

> **THE SENTENCE THAT SUMS UP THE PROGRAM**
>
> A phylogenetic tree **is not a datum: it is a hypothesis**, and like every hypothesis it is worth what its assumptions are worth. This program is built so that the assumptions are in view at all times: which model was used, with what support, with which calibration, and what would have changed with another decision. Where something cannot be known, the screen says so instead of printing a round number.

It is the sister application of *PopGeneticsPro*, which covers what happens **within** a species—allele frequencies, structure, inbreeding. PhylogenyPro starts where that one ends: above the species level, where relationships are represented by a tree and time enters the story.

I.2 The questions it answers

A program is better understood by the questions it answers than by the list of its functions. These are PhylogenyPro's, with the block that deals with each one:

Table I.1. Questions of a phylogenetic study and where they are answered.

Question	Block	What you get
Are my sequences well aligned? Which columns can I believe?	2	Built-in alignment, trimming of ambiguous blocks, quality control
Which model of evolution best describes these data?	3	56 DNA models, twelve protein matrices, Mk and Mkv, with AIC, AICc and BIC
Which is the shortest tree? And the one that best explains the data?	4 5	Parsimony, distances, maximum likelihood with NNI and SPR search

Table I.1. Questions of a phylogenetic study and where they are answered. *(continued)*

Question	Block	What you get
How much can I trust each branch?	5 6	Bootstrap, ultrafast bootstrap, SH-aLRT, aBayes, posterior probabilities
Can I reject the tree another author proposed?	5	KH, SH and AU tests on alternative topologies
Is there a molecular clock? How old are these nodes?	7	Clock test, fossil calibrations, three dating methods, chronogram
Did the group always diversify at the same pace, or was there a slowdown?	8	Lineages through time, γ, MCCR test, five models compared by AIC
What was the ancestor like? When did this character appear?	9	Mk model, ancestral states, stochastic mapping, BM/OU/early burst, PGLS
Where was the ancestor? How did it get here?	10	DEC, DIVALIKE, BAYAREALIKE with and without founder-event speciation, ancestral ranges
Why do my genes say different things?	11	Five distances between trees, gene and site concordance, networks, ABBA-BABA
How do I publish this without describing it wrongly?	12	Figure studio, export at the journal's resolution, methods written automatically

I.3 Six ideas that hold up the whole manual

These six ideas come back again and again in the twelve chapters. If you read only two pages of this manual, make it these.

1 · The alignment is a hypothesis, and it is the first one

Aligning two columns means claiming that they **descend from the same ancestral site**. It is not a preparation step: it is the first hypothesis of the study, and all the others rest on it. A badly made alignment does not produce a tree with little support, which would be honest; it produces a **firmly wrong** tree, because misaligned columns are false informative sites and they enthusiastically support groupings that do not exist. That is why Block 2 has a viewer that lets you look at the alignment column by column, and why the trimming of ambiguous blocks comes with a preview of what is going to be removed.

2 · A site that changed twice shows only the last change

Almost all the difficulty of this discipline comes from there. The differences observed between two sequences **underestimate** the changes that really happened, and they do so more and more as divergence grows, until they level off: that is **saturation**. Correcting it is what substitution models do, and choosing the model well matters as much as choosing the inference method well.

3 · More data do not fix a wrong model

It is the most expensive mistake made in phylogenetics. When the model is misspecified, adding sites does not bring you closer to the correct answer: it **increases confidence in the wrong one**. Long-branch attraction is the classic case—with infinite data, parsimony converges *more surely* to the false tree—but the logic is general and also holds for the bootstrap and for the posterior probability. A 100% support on a bad model is 100% confidence in an error.

4 · "Support" means different things depending on who says it

A bootstrap of 70% and a posterior probability of 0.70 are not the same thing, and an ultrafast bootstrap of 90% is **less** support than a classical bootstrap of 90%. Each measure has its own scale and threshold, and confusing them is the most frequent mistake in this discipline. The program has a built-in interpretation aid that, next to each number, says which band of its own scale it falls in.

5 · Time does not come from the sequences

A maximum likelihood tree has branches measured in **substitutions per site**, not in years. The only thing that turns one into the other is an external calibration: a fossil, a published rate, a geological event. And a fossil almost never gives the age of a node: it gives its **minimum age**. That is why the dates of a study are no better than its worst calibration, and why no age is ever reported without its interval.

6 · Genes that disagree are a result, not a nuisance

When two genes give different trees, the temptation is to concatenate them and move on. But the disagreement may be the interesting part: incomplete lineage sorting, hybridization, chloroplast capture, introgression. Block 11 exists to measure that conflict instead of averaging it, and a branch with 100% bootstrap and 35% gene concordance is exactly the case you need to learn to see.

1.4 What data it accepts and what it delivers

Table 1.2. Inputs and outputs of the program.

	What
Data types	DNA and RNA, amino acids, codons and discrete morphological characters.
Genomes	Chloroplast, mitochondrion, nuclear ribosomal DNA and low-copy nuclear. Each partition **declares which genome it comes from**, and that declaration is carried into all the following blocks: it is not decoration, because the chloroplast and the mitochondrion are inherited from a single parent and do not recombine.
Formats it reads	FASTA, PHYLIP (strict and relaxed, sequential and interleaved), NEXUS (DATA, CHARACTERS, SETS and TREES blocks), Clustal, MEGA, GenBank flat files and spreadsheets (.csv, .tsv) for morphological matrices. It guesses the format and the data type, and **reports everything it had to guess**.
Formats it writes	FASTA, PHYLIP, NEXUS with character sets, RAxML partition schemes, Newick, tree NEXUS with a FigTree block, phyloXML, CSV, SVG, PNG, TIFF and a .zip package with everything.
What it delivers	Trees with support, chronograms, ancestral reconstructions, figures at the resolution the journal asks for, a methods section written from what was actually run, and a reproducible package.

1.5 What you need and how to open it

A modern browser—Chrome, Edge, Firefox or Safari, from the last few years—and nothing else. You need no internet, no installation and no administrator rights.

1. **Double-click** `index.html`**.** It is the simplest way and it works for everything, because the program carries its examples embedded and does not look for any file on disk while it runs.

2. **Or with the local server**, if you prefer to open it from another computer on the same network: right-click `server.ps1` → *Run with PowerShell*, and then `http://localhost:9800` in the browser. When run as administrator, it also prints the address for connecting from a tablet on the same Wi-Fi network.

> **ABOUT SPEED**
>
> This is JavaScript, not C. One hundred maximum likelihood bootstrap replicates take minutes, and so do one hundred thousand generations of coupled Bayesian chains. The program **estimates the time before starting**, shows the progress and can be canceled. For phylogenomics with hundreds of taxa and millions of sites, keep using IQ-TREE or RAxML: the program's home page says so in those very words, and with a table of sizes that were measured, not promised.

1.6 A tour of the interface

The screen always has the same three parts.

- **The top bar.** The name of the program, the two language buttons, ES and EN, and the light or dark theme button. You can change the language and the theme whenever you like: nothing is lost or recomputed.
- **The block ribbon.** Twelve numbered steps, from 1 to 12. They are followed in order because each one uses what the previous one produced, but you can go back at any time: if you change the alignment, the blocks that depended on it will tell you.
- **The block panel.** At the top, an explanation of what is decided there and why it matters. In the middle, the controls. At the bottom, the results: tiles with the main figures, tables, plots, and a **verdict in words** that says what the result that just came out means. At the foot of each block there is a collapsible **interpretation guide** with one card per concept: what it measures, on what scale it is read and what the most common mistake with it is.

Next to many figures you will see a small circle with a question mark. Clicking it opens the card for that measure, with its scale in color bands. When the threshold the discipline uses is a custom and not a statistical law—and half of them are—the card says so.

1.7 The example data sets, and why they are simulated

The program comes with four example data sets, and all of them have a deliberate peculiarity: **the sequences are simulated on a known tree**, under a known model, with the gaps inserted lineage by lineage. The species names are real—Mexican plants, mostly of the genus *Bursera*, the copals—but the sequences are not those of any organism.

It is not a limitation: it is the reason they are good for learning. Each example carries **its own correct answer**. The true tree and the true alignment are stored alongside the data, so you can compare what you estimate with what really happened. That is impossible with real data, and it is exactly what you need when you are learning to tell a firm result from one that only looks firm.

Table 1.3. The example data sets.

Example	What it contains and what it is for
Bursera · 3 genes, unaligned	rbcL (chloroplast, coding), trnL-F (chloroplast, with indels) and ITS (nuclear ribosomal), 18 taxa. For practicing alignment and comparing the result with the true one.
Bursera · concatenated and aligned	The same three genes already aligned and concatenated, with their partition map. The starting point for Blocks 3 onwards.
Gene family · proteins	Amino-acid sequences of an MYB family. For the protein models and codon alignment.
Morphological matrix	Coded discrete characters. For Mk, Mkv and the reconstruction of ancestral states.

I.8 How a study is carried out from start to finish

The order of the blocks is not arbitrary: it is the order in which a real study is done, and each step uses what the previous one left behind.

1. **Load and align** (Block 2). Declare which genome each partition comes from. Check the alignment by eye before going on; it is the only step that no algorithm does better than someone who knows the group.
2. **Look at the assumptions before choosing** (Block 3): base composition and saturation. Then let the information criteria choose the model.
3. **Get a quick tree** (Block 4) by distances and by parsimony. It serves as a starting point and as a contrast.
4. **Do the serious inference** (Blocks 5 and 6): maximum likelihood with its support and, if time allows, Bayesian as well. That the two agree is valuable information; that they disagree, even more so.
5. **Add time** (Block 7): root, clock test, calibrations, dates with their intervals.
6. **Ask the dated tree** whatever you want to know: how the group diversified (8), how its characters evolved (9), where it was (10).
7. **Confront** (Block 11): do your genes say the same thing? And your tree and the published one?
8. **Publish** (Block 12): figure, methods and reproducible package.

Not every study goes through the twelve blocks, and it does not need to. A systematics paper may stop at 5; a biogeography one will need to go up to 10. The Block 12 report is written from what was actually run: if there was no clock, the report does not mention the clock.

I.9 How to read this manual

One chapter per block, with the same number: Chapter 7 explains Block 7. Each chapter opens with a page that says what is decided there, goes on with the minimum theory needed—no more—, then the tour of the controls, and closes with how to read the results and the frequent mistakes.

Table I.4. Typographic conventions.

Looks like this	Means
Build the tree	A button, checkbox or menu, written exactly as it appears on screen in English.
Bursera simaruba	A scientific name.
7	A cross-reference to another block and its chapter.

THE FOUR KINDS OF BOX

Note, with information that helps. **Idea**, with something worth remembering beyond the program. **Warning**, with a caution that can be costly if ignored. And **Frequent mistake**, with a specific error that is often made and how to avoid it.

I.10 What the program does not do

An honest manual also says where its program ends. PhylogenyPro does **not** do any of this:

- **Large-scale phylogenomics.** Hundreds of taxa and millions of sites are out of its reach because of speed, not method. The program's home page has a table of sizes **measured** on an ordinary desktop computer.
- **Full multispecies coalescent.** There is a species tree from quartets, which is a heuristic, not the exact ASTRAL algorithm nor a coalescent model with a likelihood.
- **Explicit phylogenetic networks with hybridization.** There are split networks, which **show** the conflict without explaining it; there is no estimation of a number of hybridization events.
- **Annotation or sequence searching.** It does not connect to GenBank or to any database: it has no network access, by design.
- **Deciding for you.** The program computes, compares and warns; the interpretation, the choice of calibrations and the definition of the areas remain yours, and they are what you will have to defend before a reviewer.

BEFORE MOVING ON TO CHAPTER 1

If some of the above seems overly cautious, it is worth remembering where the caution comes from. Molecular phylogenetics has spent forty years discovering that its firmest certainties—the 100% support, the date with three decimals, the ancestor reconstructed with confidence—were often artifacts of an assumption that nobody had checked. This program is written with that memory on its shoulders.

1

CHAPTER ONE · BLOCK 1

The home page: what to believe before you start

The first block does not compute your tree: it prepares you to compute it. It brings the complete route of the study, the catalog of what the application can do, a frank comparison of the four inference criteria, a table of **measured** sizes and—most important—two labs that compute real phylogenetics on the page, so that two of the hardest ideas in the discipline stop being a warning and become something you saw happen.

1.1What a home page that computes is for

1.2The route of the study in seven steps

1.3Lab 1: the Felsenstein zone

1.4Lab 2: saturation and correction

1.5What you can analyze, and with which marker

1.6The four criteria, compared plainly

1.7How far it goes, and how far it does not

1.8The method gallery and the references

1.9The nine theory sections

1.10Frequent mistakes

1

1.1 What a home page that computes is for

Almost every home page of a scientific program is a brochure: it says what the program does and invites you to begin. This one does something else. Besides presenting the route, it **computes**: two labs run real phylogenetics—exact probabilities, simulation, inference by three methods—every time you move a control. Nothing you see there is an animation.

The reason for that design is pedagogical, and it has to do with idea 3 of the introduction. You can warn a thousand times that "more data do not fix a wrong model," and the warning slides off. Seeing, on your own screen, how parsimony recovers the false tree **more often as you increase the number of sites** does not slide off: that is not forgotten.

> **WHAT IS DECIDED IN THIS BLOCK**
>
> Nothing, technically: it produces no result that the other blocks use. What is decided here is **what you expect from the study you are about to do** and what you are going to believe from it. It is the only block worth reading in full before loading data, and the only one worth coming back to when a result does not add up.

The six cards in the header sum up the conditions you will work under: everything runs in the browser, your data never leave your computer, it accepts DNA, proteins, codons and morphology, it treats the chloroplast, the mitochondrion and the nucleus separately, it is in Spanish and English, and it exports figures at up to 900 dpi.

1.2 The route of the study in seven steps

The "How it works" strip groups the twelve blocks into the seven steps of a real phylogenetic study, in the order in which it is done:

Table 1.1. The seven steps of the strip and the blocks that carry them out.

Step	Called	What happens	Blocks
1	**Align**	Sequences, trimming of ambiguous blocks, partitions	2
2	**Choose a model**	Saturation, AIC and BIC	3
3	**Infer**	Parsimony, maximum likelihood and Bayesian	4 5 6
4	**Measure support**	Bootstrap and posterior probabilities	5 6
5	**Date it**	Clock, fossils, geological scale	7
6	**Interpret**	Traits, areas, diversification	8 9 10 11
7	**Publish**	Figure, report and ZIP package	12

Below, the twelve block cards say what each one does. **Each card corresponds to a chapter of this manual, with the same number.** Clicking a card jumps to the block; if its requirements are not ready, the block will say so instead of computing something meaningless.

1.3 Lab 1: the Felsenstein zone

This is probably the most important corner of the whole program. It illustrates the result of Felsenstein (1978) that founded the modern critique of parsimony, and it does so by computing, not by telling.

What it shows

A tree of **four taxa** with the shape that does harm: two long branches that are **not** sisters, separated by short branches. With four taxa there are only three possible unrooted topologies, and with four possible states per site there are 256 patterns; but by symmetry those 256 reduce to the **15 partitions of the set {A, B, C, D}**, so the probability of each pattern under Jukes–Cantor can be computed **exactly**, without simulating anything.

The panel uses that exactness to answer the question with infinite data: which informative pattern is the most probable? If the pattern that gathers the most probability is the one that groups the two long branches, then parsimony, given enough sites, will choose **the wrong tree with complete certainty**. That is what being *inconsistent* means: converging to the false answer.

The controls

- Long branches and Short branches: the two lengths that define the scenario, in substitutions per site.
- Sites: how many sites each simulated alignment has.
- Replicates: how many alignments are simulated to measure how often each method gets it right.
- ↻ New replicates: simulates again with another seed.

What to read

Three things, in this order:

1. **The true tree** and **Where parsimony fails**: the panel computes by bisection the **boundary** of the scenario, that is, the exact point where the long branch becomes too long and parsimony starts getting it wrong. Moving the sliders to one side and the other of that boundary is the whole lesson.
2. **The three informative patterns** and their exact probability. When the "long branches together" pattern overtakes the true pattern, that is it: parsimony has lost, however much data you give it.
3. **How often each method recovered the true tree**, over the simulated replicates, with parsimony, with neighbor joining on Jukes–Cantor-corrected distances, and with maximum likelihood optimizing the branch lengths on each of the three topologies.

> **THE EXPERIMENT YOU MUST DO**
>
> Set up a scenario inside the zone—long branches of 0.6 and short ones of 0.03, for example—and then **raise the sites** from 100 to 2,000. Likelihood improves, as you would expect. Parsimony **gets worse**: with more data it is right less often, because each new site reinforces the false signal. That is the difference between a consistent method and one that is not, and it is idea 3 of the introduction seen live.

The "Why this matters in your own data" panel translates it: a genus with one fast-evolving species, a very distant outgroup or a saturated marker produce exactly this geometry without anyone having looked for it. If your tree joins the outgroup with the most divergent taxon, suspect this before you suspect the biology.

1.4 Lab 2: saturation and distance correction

The second lab makes idea 2 visible: a site that changed several times shows only the last change.

Two sequences diverge under Kimura's two-parameter model, with rates that vary among sites according to a gamma distribution. The panel plots the **estimated distance against the true distance**, with three overlaid curves: the **raw differences** (the proportion of differing positions), the model correction, and the line of the **perfect estimator**, which is the diagonal.

What you see is immediate: the raw differences start out stuck to the diagonal and peel away more and more, until they flatten out. That is saturation. The correction brings them back to the diagonal, but only as far as it has a solution; beyond that, the formula breaks down and the panel shows it.

The controls

- `Maximum divergence`: how far the two sequences are allowed to drift apart.
- `ts/tv ratio`: how many transitions there are for each transversion. Raise it and you will see that transitions saturate much earlier.
- `Γ shape`: the inequality of rates among sites. Low values—a few very fast sites and many almost frozen—bring saturation forward.
- `Sites`: how many positions each sequence has; with few, noise dominates.
- `rate variation among sites (+Γ)`: turns the heterogeneity off and on to see how much it changes.

The second plot, **Transitions and transversions**, is the one you will use with your own data in Block 3. The rule can be read straight off the screen: **when the transition curve stops rising and starts to fall, the marker is saturated**. It is not a metaphor: accumulated transitions are lost because the sites change again.

SATURATED DOES NOT MEAN USELESS

A saturated marker still resolves recent nodes well; what you cannot do is use it for the deep nodes **and believe its support**. The usual solution is not to throw the gene away: it is to remove the third codon position, switch to amino acids, or look for a slower marker for the deep part of the tree.

1.5 What you can analyze, and with which marker

The "What you can analyze" section lists the data types that the same likelihood engine handles: DNA, proteins, codons and morphology. They are not four programs inside, but one that works with four, twenty or however many states are needed.

Right after comes the list of markers, and it is worth pausing there, because it is the one that prevents the most expensive mistake in a plant study: **treating all genes as if they told the same story**.

Table 1.2. The markers the application recognizes by name, with their genome and their character.

Genome	Markers	What to expect
Chloroplast	rbcL	The workhorse: slow, easy to amplify, good from family level upwards.
Chloroplast	matK	More variable than rbcL; the other plant barcode marker.
Chloroplast	trnL-F	Non-coding spacer: resolves between close species.
Chloroplast	trnH-psbA	The most variable chloroplast spacer; full of indels.
Chloroplast	ndhF, atpB, rpl16, rps16, ycf1	The rest of the usual repertoire in plant systematics.
Nuclear ribosomal	ITS, ITS2, ETS	Fast and widely used, but with divergent copies and incomplete concerted evolution.
Low-copy nuclear	waxy, PHYC, LEAFY, G3pdh	Biparental: they are the ones that **detect hybridization**.
Mitochondrion	matR, nad1, cox1, atp1	The slowest in the plant; useful in deep phylogenies.

WHY THE GENOME IS NOT DECORATION

The chloroplast and the mitochondrion of a plant are inherited **from a single parent**—in most angiosperms, from the mother—and **do not recombine**: the whole genome is a single lineage, with a single history. The nucleus is inherited from both and does recombine. That is why a chloroplast tree and a nuclear tree **can disagree without either of them being wrong**: chloroplast capture after an ancient hybridization leaves exactly that signature. Declaring the genome of each partition in Block 2 is what lets the following blocks treat that disagreement as information and not as noise.

1.6 The four criteria, compared plainly

The comparison table on the home page is deliberately frank: **no method wins every time**. This is its printed version.

Table 1.3. The four inference criteria, what they optimize and where they break.

	Distances (NJ)	Parsimony	Maximum likelihood	Bayesian
What it optimizes	Fit to a distance matrix	The fewest changes	The probability of the data given the tree	The probability of the tree given the data
Uses a substitution model	Yes, when correcting the distances	No (or only weights)	Yes, explicit	Yes, explicit and with priors
Consistent with unequal long branches	Depends on the correction	**No: the Felsenstein zone**	Yes, under the correct model	Yes, under the correct model
Measure of uncertainty	Bootstrap	Bootstrap, jackknife, Bremer	Bootstrap, UFBoot, SH-aLRT	Posterior probability
Estimates branch lengths in substitutions	Yes	In steps, not substitutions	Yes	Yes, with an interval
Allows the tree to be dated	Only with approximate methods	No	Yes (clock, LSD)	Yes, with credibility intervals
Computational cost	Seconds	Seconds to minutes	Minutes	Minutes to hours
What it is best for	A first look and the starting tree	Morphology and data with few changes	**The tree that gets published**	Uncertainty, dates and complex models

Note that the decisive row is the third, and that it is exactly what Lab 1 lets you see. Parsimony is not "worse": it is **inconsistent under a specific geometry**, and it remains the right tool for morphology, where no substitution model is credible and changes are few.

1.7 How far it goes, and how far it does not

The table of limits on the home page is not a sales pitch: these are **times measured** on an ordinary desktop computer. It is worth looking at before loading a large matrix.

Table 1.4. Sizes the application handles, measured. "Comfortable" runs without a second thought; "with a warning" can still be done, with patience.

Analysis	Comfortable (taxa)	With a warning	Typical time
Distances, NJ, parsimony	≤ 300	≤ 1,000	Seconds
Likelihood + ultrafast bootstrap	≤ 120	≤ 300	Minutes

Table 1.4. Sizes the application handles, measured. "Comfortable" runs without a second thought; "with a warning" can still be done, with patience. *(continued)*

Analysis	Comfortable (taxa)	With a warning	Typical time
Likelihood + classical bootstrap	≤ 60	≤ 120	Minutes with several threads
Bayesian (MC³)	≤ 50	≤ 100	Minutes to hours
Least-squares dating	≤ 500	≤ 1,000	Seconds
Bayesian dating	≤ 60	≤ 120	Minutes to hours
DEC biogeography	≤ 8 areas	≤ 10 areas	Minutes

The biogeography limit is different from the rest and deserves a note: it depends not on the number of taxa but on the number of **areas**, because the number of possible ranges grows as 2 raised to the number of areas. With eight areas there are already 256 states; with twelve, more than four thousand.

Below the table, the home page says in one line what needs to be said: for phylogenomics with hundreds of taxa and millions of sites, keep using IQ-TREE or RAxML. This is JavaScript, and no optimization closes the gap with C.

1.8 The method gallery and the references

The gallery lists **everything** the application computes, grouped into eight families—data, models, quick trees, inference, time, macroevolution, biogeography, and compare and publish—with a filter to see a single family. It serves two purposes: checking at a glance whether the method you need is there, and finding which block it lives in.

At the end of the home page there are **94 references**, also grouped by family. They are not there for decoration: each method the application implements cites the article in which it was published, and the Block 12 report builds its bibliography by taking from that same list **only what was actually used** in your run.

> **I** CITE THE METHOD, NOT JUST THE PROGRAM
>
> When you publish, cite the program **and** the original articles of the methods you used. An experienced reviewer will notice the difference between "program X was used" and "a GTR+Γ model (Yang 1994) was fitted with SPR search and ultrafast bootstrap support (Minh et al. 2013)." The methods section that Block 12 writes already comes with those citations in place.

1.9 The nine theory sections

The home page closes with nine collapsible sections that are, in practice, a short course. They are worth it even if you already know the subject, because they fix the vocabulary that the rest of the program uses.

1. **What a phylogenetic tree is (and what it is not).** Topology, branch lengths, root; why an unrooted tree has no "ancestors."
2. **The alignment is a hypothesis of homology.** Idea 1 of the introduction, developed.
3. **The changes you cannot see: substitution models.** Why correcting is needed, and what each model corrects.
4. **Parsimony, likelihood and Bayes: how they really differ.** Beyond the slogan of "the simplest tree."
5. **What a support value actually means.** Idea 4, with the scales of each measure.
6. **The molecular clock and calibrations.** Idea 5.
7. **Chloroplast, mitochondrion and nucleus: three histories in one organism.** What makes Block 11 exist.
8. **Ancestral areas: how a biogeographic history is reconstructed.** The framework of Block 10.
9. **The eight mistakes that show up most in a manuscript.** The section worth reading *before* writing, not after the paper is rejected.

1.10 Frequent mistakes

SKIPPING THE HOME PAGE

It is the mistake everyone makes. The two labs take ten minutes and change the way you look at the results of the eleven blocks that follow. If you only have time for one, make it the Felsenstein zone.

READING THE TABLE OF LIMITS AS A PROMISE

The times were measured on a specific computer with a specific alignment. The table tells you whether you are in the right order of magnitude, not when to set a deadline.

CONCLUDING THAT "PARSIMONY IS BAD"

Lab 1 does not show that. It shows that parsimony is **inconsistent under a specific geometry** of branches. Outside that geometry it behaves well and, for morphological matrices—where there is no credible substitution model—it remains the right tool. Chapter 4 returns to this.

CHOOSING THE MARKER BY LAB HABIT

"We always use ITS" is not an argument: ITS resolves between close species, but at a deep node it will give a confident and wrong answer. The home page says at what depth each marker is informative.

2

CHAPTER TWO · BLOCK 2

Data and alignment

The most important block in the program and the one that usually gets the least attention. Here the sequences are read, aligned, checked **by hand**, the doubtful columns are trimmed and the partition map is built, declaring which genome each partition comes from. Everything the following ten blocks do rests on what is decided in this one, and no later method can correct a mistake made here.

2.1Why this block decides the study

2.2Loading the data: formats and what is guessed

2.3Partitions: the map of the study

2.4Aligning: the aligner and its options

2.5Aligning by codon

2.6The viewer: really looking at the alignment

2.7Editing by hand

2.8Trimming of ambiguous blocks

2.9Quality control: eleven things that are checked

2.10Exporting, and what goes on to Block 3

2.11Frequent mistakes

2

2.1 Why this block decides the study

Aligning two columns means claiming that they **descend from the same ancestral site**. It is not data preparation: it is the first hypothesis of the work, and it is the only one that no later block can put to the test. The maximum likelihood of Block 5 will take your alignment as a fact and compute on it in all seriousness; if the columns are wrong, it will compute a false answer in all seriousness.

> **THE CONSEQUENCE YOU NEED TO UNDERSTAND**
>
> A bad alignment **does not produce a tree with little support**, which would be the honest result. It produces a **firmly wrong** tree: the misaligned columns become false informative sites, and those sites enthusiastically support groupings that do not exist. The bootstrap will back them, because the bootstrap measures the consistency of the signal, and the signal is there—it is just artificial.

That is why this block does not stop at aligning: it brings a viewer to look at the result column by column, an editor to correct it by hand, a trimming step with a preview and a quality control that looks for eleven specific problems. The sentence at the head of the alignment section on screen says it plainly: *the eye of someone who knows the group often beats the algorithm*.

2.2 Loading the data: formats and what is guessed

Each file you load comes in as a **partition**: a gene, a region or a matrix. You can load several at once, by dragging them onto the box, choosing them with Choose files... or pasting the text with Paste sequences.

Table 2.1. Formats the program reads, and what you need to know about each.

Format	Detail
FASTA	The simplest and the safest. If in doubt, convert to FASTA.
PHYLIP	Strict and relaxed, sequential and interleaved. The strict one cuts names at ten characters; the program detects which is which and says so.
NEXUS	DATA, CHARACTERS, SETS (with *charsets*, which become partitions) and TREES blocks. It also understands MATCHCHAR.
Clustal	The classic output of ClustalW and ClustalΩ.
MEGA	The MEGA format, with its `#MEGA` headers.
GenBank	Flat files. Names are taken from the ORGANISM field, and from the gene when it is present.
Spreadsheet	.csv and .tsv, for morphological matrices: one row per taxon, one column per character.

> **WHAT IT GUESSES, IT SAYS**
>
> The program deduces the format and the data type—DNA, protein or morphology—and **reports everything it had to guess**. When a GenBank file forces it to take the names from the ORGANISM field, it warns you and asks you to check them before going on: that is where duplicate names and synonyms sneak in.

If you prefer to start without your own data, the four example buttons load the data sets described in Section I.7: `Bursera · 3 genes, unaligned`, `Bursera · concatenated and aligned`, `Gene family · proteins` and `Morphological matrix`. The three unaligned genes are the right choice for practicing this whole chapter, because they carry their true alignment stored with them and let you check how well you did.

2.3 Partitions: the map of the study

The list of partitions is the map of what you are going to analyze. Each row shows the name, how many taxa and positions it has, whether it is aligned, and two menus: the **genome** and the **data type**.

> **THE GENOME IS NOT DECORATION**
>
> Declaring whether a partition comes from the **chloroplast**, the **mitochondrion**, **nuclear ribosomal DNA** or the **low-copy nucleus** is the decision in this block that reaches furthest. The chloroplast and the mitochondrion are inherited from a single parent and do not recombine: the whole genome is one lineage with a single history. The nucleus is inherited from both and recombines. That declaration travels to all the following blocks, and it is what allows Block 11 to treat a disagreement between genes as **information**—chloroplast capture, hybridization—and not as noise to be averaged away.

Three buttons act on the whole set:

- `Concatenate all` joins the aligned partitions into a single matrix, filling with missing data the taxa that are absent from any of them. The concatenated matrix keeps the map of where each gene starts and ends, which is what Block 3 will use for the partition scheme. You need **at least two aligned partitions**.
- `Split by codon position` divides a coding gene into three partitions—first, second and third position—, which is what is done when the third is saturated and you want to model it separately or discard it.
- `Import a partition scheme...` reads a scheme already written, in the style of RAxML's.

2.4 Aligning: the aligner and its options

The aligner is progressive, from the school that Clustal made standard and MUSCLE refined. It works in four steps:

1. **Quick distances** between each pair of sequences from the *k*-mers they share, without aligning anything yet.
2. **Guide tree** by UPGMA on those distances.
3. **Profile-to-profile alignment** traveling the guide tree from the leaves to the root, with affine gaps (Gotoh 1982) and with the penalty reduced where the profile already has gaps.
4. **Iterative refinement**, optional: each branch of the guide tree splits the sequences in two, the two sub-alignments are realigned and the result is kept **only if it improves** the sum-of-pairs score. It is the third stage of MUSCLE.

There are four controls: `by codon`, `refine`, `gap open` and `extend`. The last two are Gotoh's penalties: opening a gap costs a lot, extending it costs little, because an insertion of six bases is a single event and not six.

> **HOW GOOD IT IS**
>
> Measured against the true alignment of the simulated examples, the aligner recovers **99.98%** of the pairs in rbcL, **98.14%** in trnL-F—the one that carries indels—, **98.93%** in ITS and **99.60%** in the protein family. Aligning rbcL **by codon**, **100%**. Those figures are in the validation folder and can be reproduced.

2.5 Aligning by codon

If the partition is coding, this is the most useful checkbox in the whole block. With `by codon` checked, the program translates, aligns as protein and returns the gaps to the nucleotides **three at a time**.

The consequence is that **a gap of one or two bases cannot appear**, and such a gap, in a coding region, is almost always an artifact of the aligner: it would imply a frameshift that selection would have removed. Aligning by codon does not improve the result a little: it eliminates an entire class of error at the root, and that is why the rbcL example goes from 99.98% to 100%.

2.6 The viewer: really looking at the alignment

The viewer draws the alignment on a canvas and copes with thousands of columns without slowing down. Three menus change what you see:

- `color`: `by residue` (each base or amino acid with its own), `differences from consensus` —which is the view that gives away a misaligned column, because the differences bunch up—or `no color`.
- `mark variable and informative`: flags which columns change and which are parsimony-informative. Remember that the informative ones are the ones that choose the topology; the rest do not.
- `codon positions`: marks first, second and third. It shows at a glance whether the third is a mess, which is the signature of saturation.

TEN MINUTES HERE ARE WORTH MORE THAN ANYTHING ELSE

Go through the alignment from beginning to end with the coloring by differences from consensus. Look for regions where a single sequence deviates completely—usually a sequence in the wrong orientation, or from another gene—, blocks where the gaps are staggered for no reason, and ends with data from a single taxon. They are minutes that save weeks.

2.7 Editing by hand

The editor has **unlimited undo and redo**, which is what makes it safe to experiment. There are eight operations:

Table 2.2. The editor's operations.

Operation	What it does
Gap in this sequence	Inserts a gap at the marked position, in that sequence only.
Gap in all	Inserts a column of gaps in all of them: it shifts the alignment without breaking it.
Remove this gap	Closes a gap in one sequence.
Delete column(s)	Deletes the selected columns.
Drop empty columns	Deletes the ones left without any residue, which tend to appear after editing.
Sort by name	Reorders the rows alphabetically, to find a taxon.
Remove taxon	Takes a whole sequence out of the partition.
↶ Undo / ↷ Redo	With no limit on the number of steps.

Editing by hand is not cheating: it is what any experienced systematist does, because an algorithm does not know that this region is a microsatellite or that this taxon has a known duplication. What is mandatory is **saying so in the methods**, and the Block 12 report records that the alignment was edited.

2.8 Trimming of ambiguous blocks

There are regions—highly variable spacers, poorly sequenced ends—where the alignment is simply not credible. Removing them reduces noise; removing too much erases exactly what is informative. The program offers two schools:

Gblocks (Castresana 2000)

It chooses by **neighborhood** rules: a column is kept if it and its neighbors meet certain requirements. Its parameters are conserved and highly conserved (how many sequences must agree), max contiguous (how many non-conserved columns in a row are tolerated), minimum block (the minimum length of a block that is kept) and gaps allowed, with three levels: none, up to half and all.

Per-column thresholds, in the style of trimAl

They decide column by column: by gap proportion, by column similarity, or gaps and similarity at once, with the max gaps and min similarity sliders.

Whatever the method, the program **first shows what it is going to remove**. The preview paints the doomed columns on the alignment, and nothing is touched until you click Apply the trimming; Clear the preview cancels. The keep whole codons checkbox forces whole codons to be removed, so as not to break the reading frame.

> **ALWAYS TRIMMING, OUT OF HABIT**
>
> Trimming does not always improve the tree. There are published studies where removing columns **worsens** the inference, because informative sites go out together with the noise. The honest check is to run the analysis **with and without trimming** and see whether the tree changes. If you lost more than half of the columns, the problem was not the noise: it was the alignment.

2.9 Quality control: eleven things that are checked

The last section of the block looks for "what ruins an analysis without ever showing up in a spreadsheet." These are the checks, with what each warning means:

Table 2.3. The quality-control checks.

Check	Why it matters
Repeated names	Two taxa with the same name break any later analysis, and the error shows up much later.
Names with awkward characters	Brackets, commas and accents that other programs reject: it matters if you are going to export to NEXUS or Newick.
Very short sequences	Those less than half the length of the longest will hang from almost nothing and their position will be unstable.
Identical sequences	Not an error, but their branches end up at zero and some analyses misbehave. It is worth knowing beforehand.
Unexpected characters	Letters that do not match the declared data type: almost always, a protein marked as DNA or vice versa.
Taxa with many gaps	More than half gaps or missing data in a taxon.

Table 2.3. The quality-control checks. *(continued)*

Check	Why it matters
Empty columns	With no residue at all; they tend to be left after editing and should be removed.
How many columns are constant	The proportion of columns that do not change, and how many are variable. It is the real raw material of the analysis.
Internal stop codons	In a coding region they indicate a wrong reading frame or a pseudogene. The program tries the three frames and says which is the right one.
Length a multiple of three	If it is not, the last codon is incomplete.
Unequal G+C between sequences	The subtlest warning and the most important: unequal composition **breaks the assumption** of almost all models, and makes two unrelated taxa attract each other in the tree. Block 3 puts it to a formal test.

✦ THE TWO WARNINGS NEVER TO IGNORE

Internal stop codons and **unequal G+C composition.** The first almost always means something is wrong in the data—wrong frame, reversed sequence, pseudogene—and is easy to fix. The second is not fixed by changing substitution model: it is dealt with by removing the third codon position, switching to amino acids or recoding, and Chapter 3 explains why.

2.10 Exporting, and what goes on to Block 3

The block writes FASTA, PHYLIP, NEXUS with its *charsets*, RAxML-style partition schemes and CSV. It serves both to save the work and to take it to another program.

What goes on to Block 3 is the set of aligned partitions, each with its **genome**, its **data type** and—if it is coding—its reading frame. If you come back to this block and change the alignment, the blocks that depended on it will tell you instead of going on showing an old result.

2.11 Frequent mistakes

✕ ALIGNING AND MOVING ON TO THE NEXT BLOCK WITHOUT LOOKING

It is the most expensive mistake in the whole program, and the most common. The alignment is the only hypothesis that no later block can put to the test. Ten minutes in the viewer are worth more than any refinement of the model.

✕ NOT ALIGNING A CODING REGION BY CODON

It costs one checkbox and eliminates an entire class of error. If your partition is coding and you aligned it by nucleotide, go back and do it again.

LEAVING THE GENOME AT ITS DEFAULT VALUE

The genome menus start with a guess based on the marker's name. If your partition is called "gene1," that guess will be bad. Correct it: Block 11 depends on it to tell a real biological conflict from an error.

CONCATENATING BEFORE LOOKING AT EACH GENE SEPARATELY

Concatenation averages. If two genes have different histories—which is exactly what this program can detect in Block 11—, concatenating first hides the finding. Align and check each partition on its own, and concatenate afterwards.

CONFUSING "MANY POSITIONS" WITH "MUCH DATA"

A thousand positions of which thirty are parsimony-informative resolve little. The quality control says how many columns are constant and how many variable; that is the figure to look at, and the one worth reporting in the paper.

3

CHAPTER THREE · BLOCK 3

Substitution models

A site that changed twice shows only the last change. This block deals with that: first it tests the assumptions that every model takes for granted—homogeneous composition and an unsaturated signal—, and then it fits 56 DNA models, twelve protein matrices or Mk and Mkv for morphology, and lets the information criteria choose. It is not a formality: the chosen model determines the branch lengths, the support and, in difficult cases, the topology itself.

3.1What is decided in this block

3.2What your data contain

3.3The assumptions, put to the test

3.4The catalog of models

3.5How the choice is made: AIC, AICc, BIC and weights

3.6Reading the winning model

3.7Hierarchical likelihood ratio tests

3.8The partition scheme

3.9Frequent mistakes

3.1 What is decided in this block

The differences observed between two sequences **underestimate** the changes that really happened, because a site that changed from A to G and then to T is counted as one change, not two, and if it went back to A none is counted at all. A substitution model is the tool that turns what is observed into an estimate of what happened.

Choosing it well matters as much as choosing the inference method. A model that is too simple underestimates long branches—and thereby feeds the long-branch attraction of Chapter 1—; one that is too complex spends parameters the data cannot estimate and increases variance without removing bias.

> **WHY "ALWAYS USE GTR+I+G" IS NOT ENOUGH**
>
> It is what many people do, and with thousands of sites it rarely goes wrong. But with little data GTR+I+G has eight free parameters plus the branch lengths, and estimating them all on two hundred positions produces very imprecise estimates. Besides, **+I and +G compete to explain the same thing**: together they tend to give extreme values of both that do not mean what they seem to. This block exists so that the choice is made by fit and not by habit.

The block works on **one aligned partition** at a time, chosen in the partition menu. If you have several, section 4 of the panel also looks for how to group them.

3.2 What your data contain

The first section summarizes the partition before touching anything: how many taxa, how many sites, how many columns are **constant**, how many **variable** and how many **informative**, the G+C content and the proportion of gaps.

The sentence at the head of the section says something worth understanding well: *constant sites tell you nothing about the topology but plenty about the model*. A column where all taxa have A does not choose between topologies, but it does say that there were no changes at that site, and that is direct information about the rates and about the proportion of invariable sites. That is why **removing constant sites "because they contribute nothing" biases branch lengths upward**: the model believes there was more change than there was.

3.3 The assumptions, put to the test

Before choosing a model, the block checks the two assumptions that **no** model on the list relaxes.

Composition per sequence

All the usual models assume that base composition is the same in every sequence and throughout the tree. The plot shows the composition of each sequence, and a χ^2 test compares each one against the mean.

> **! WHAT TO DO IF COMPOSITION IS HETEROGENEOUS**
>
> What does **not** work is changing the substitution model: none of the 56 fixes it, because they all share that assumption. Two sequences with similar composition attract each other in the tree even if they are not related, and the support will be high. There are three real ways out: **removing the third codon position**, which is usually the culprit; **switching to amino acids**, whose composition varies less; or **recoding** the nucleotides into purines and pyrimidines. The help card of the χ^2 test says so on screen as well.

Saturation

The second plot is the one you practiced in Lab 2 of Chapter 1, now on your data: transitions and transversions against the corrected distance. The rule reads the same way: **as long as the cloud follows a straight line, the marker is informative; when it bends and flattens, it is saturated**. Transitions saturate before transversions, and the third codon position before the first and second.

Remember what saturated means and what it does not mean: the marker still resolves recent nodes well; what it cannot do is support the deep nodes **and their support values**.

3.4 The catalog of models

For DNA, the catalog is built by combining three things: seven **rate schemes**, two ways of treating the **base frequencies** and four **among-site rate variation variants**. Seven times two times four makes **56 models**.

Table 3.1. The seven rate schemes. The name on the left is the one it carries with equal frequencies; the one on the right, with empirical frequencies.

Equal freq.	Empirical freq.	Free rates	What it distinguishes
JC	F81	0	All changes at the same rate.
K80	HKY	1	Transitions and transversions at different rates: the parameter κ.
TrNef	TrN	2	The two classes of transition are separated.
K81	K81uf	2	Two classes of transversion.
TIMef	TIM	3	Separate transitions and separate transversions.
TVMef	TVM	4	All transversions free, transitions equal.
SYM	**GTR**	5	The six rates free: the most general.

And the four variants, which apply to any of the fourteen:

Table 3.2. The four variants of among-site rate variation.

Variant	What it adds
(none)	All sites change at the same speed. It is almost never true.
+I	A proportion of sites **unable** to change, not simply sites that did not change.
+G	Rates are distributed according to a discrete gamma (Yang 1994), with shape α. It is the variant that usually matters most.
+I+G	Both at once. They compete to explain the same thing; see the warning in Section 3.6.

For **proteins**, the catalog is twelve empirical matrices—LG, WAG, JTT, Dayhoff, cpREV, mtREV24, VT, RtREV, Blosum62, FLU, HIVb and mtmam—, also with the four variants. The `quick` set tries six (LG, WAG, JTT, cpREV, Dayhoff and VT) and the `standard` set all twelve. **cpREV is there for a specific reason**: it was estimated from chloroplast-encoded proteins, which is where most markers in plant systematics come from.

> **WHERE THOSE TWELVE MATRICES COME FROM**
>
> They are the matrices published in their original articles, each one cited in the code next to its numbers, and checked value by value against five independent canonical distributions—PAML, IQ-TREE, MrBayes, RAxML and the file that the WAG authors themselves deposited at the EBI. The `PROCEDENCIA.md` file documents the check.

For **morphology**, the model is **Mk** (Lewis 2001): a *k*-state model with all rates equal. Its variant **Mkv** adds the correction for the fact that morphological matrices almost never include constant characters—nobody codes the character "has cells"—, and without that correction branches come out inflated.

The controls

- `set`: `quick` or `standard`, depending on how many candidates you want to try.
- `criterion`: AIC, AICc or BIC, which is the one that sorts the table.
- `Γ categories`: how many discrete classes the gamma is split into. Four is the usual number and what almost everyone reports.
- `starting tree`: `BIONJ, ML-optimized`, `neighbor joining` or `UPGMA`. All candidates are fitted **on the same tree**, which is what makes the comparison fair.

3.5 How the choice is made: AIC, AICc, BIC and weights

A model with more parameters **always** gives a better likelihood, even if the extra parameters are of no use at all. That is why the lnL is not compared on its own, but a quantity that rewards fit and penalizes parameters:

THE THREE FORMULAS

AIC = $-2\cdot\ln L + 2k$ · **AICc** = $\mathrm{AIC} + 2k(k+1)/(n-k-1)$ · **BIC** = $-2\cdot\ln L + k\cdot\ln(n)$

where k is the number of parameters and n the number of sites. In all three, **lower is better**. AICc corrects for small samples and approaches AIC when n is large. BIC penalizes more heavily and therefore tends to choose simpler models.

What you read in the table is not the absolute value but the **difference from the best**, ΔAIC. Burnham and Anderson's convention: below 2, the model is practically indistinguishable from the winner; between 4 and 10, clearly worse; above 10, discarded.

The **Akaike weights** column turns those differences into probabilities that sum to one: the weight of a model is the probability that it is the best **of the set compared**. A winner with a weight of 0.9 settles the matter; several models sharing the weight means that the data do not distinguish between them, and then it is worth checking that the tree does not change depending on which one is used.

"THE BEST OF THE SET" IS NOT "GOOD"

If all the candidates are bad for these data—for example, because composition is heterogeneous and none of them models it—, the weight of 0.99 goes to the least bad. Choosing the model is not the analysis, and a GTR+Γ fitted on a crooked alignment is still a GTR+Γ fitted on a crooked alignment.

3.6 Reading the winning model

Once the model is chosen, the program **refits it properly**, with all branches optimized, and shows its parts.

The rate matrix

The six exchangeability rates between pairs of bases and the four equilibrium frequencies. It is worth looking at the ratio between transitions and transversions: in nuclear and chloroplast DNA the normal value is around 2, and a value close to 1 in a gene that should have more is a sign of saturation—the transitions, which saturate first, have already been erased.

The shape of the gamma (α)

It describes how unequal the distribution of rates among sites is. Below 1 the heterogeneity is strong: a few very fast sites and many almost frozen. Above 3, rates are almost equal and +G adds little. The **Rates among sites** plot draws the discrete categories with their relative rate.

> **INTERPRETING A AND THE PROPORTION OF INVARIABLE SITES SEPARATELY IN A +I+G**
>
> +I and +G explain the same phenomenon in two ways, and when they are fitted together they share the work unstably: it is common to get a high proportion of invariable sites **and** a low α, with neither value meaning what it seems to. Many authors recommend using +G alone. If you need to report I and α, do so stating that they come from a joint fit.

The verdict in words

Below the tables, the panel writes in prose which model won, by how much, and what its parameters mean. It is the text you can take almost as is into the methods section, and the one Block 12 reuses when it writes the report.

3.7 Hierarchical likelihood ratio tests

The other way of choosing, older than the information criteria and still often requested by reviewers. It compares **nested** models—one is a special case of the other—with the statistic $2\Delta\ln L$ against a χ^2 with as many degrees of freedom as the difference in parameters, and chains the tests until it settles on the simplest model that is not rejected.

The program offers it because it is still asked for, and at the same time it says why information criteria are the preferred route today:

- The result **depends on the order** in which the tests are chained, and there is no canonical order.
- It only compares nested models, so it leaves out perfectly sensible comparisons.
- The χ^2 is an asymptotic approximation and **does not hold when a parameter sits on the boundary of its range**, which is exactly the case of +I: the proportion of invariable sites is tested against zero, which is its limit.

3.8 The partition scheme

With several partitions the question is not only which model, but **how many different models are needed**. Giving each gene its own is the most flexible and the most expensive in parameters; giving all of them a single one is the cheapest and may be false.

Find the scheme carries out a **greedy merging** in the style of PartitionFinder: it starts with one partition per gene, at each step tries the merger of two that improves the information criterion, and stops when no merger improves it. The result says which partitions share a model and which deserve their own.

With a gene split by codon position, the typical result is that the first and second positions merge and the third stays on its own, which is exactly what biology predicts.

The Export the chosen model button saves the scheme and the model to take them to another program or to keep a record.

3.9 Frequent mistakes

CHOOSING THE MODEL WITHOUT FIRST LOOKING AT THE ASSUMPTIONS

The two plots in section 2 of the panel are more important than the whole table of models. If composition is heterogeneous or the marker is saturated, the winning model will be the least bad of a list of unsuitable models, and its Akaike weight of 0.99 will say nothing about it.

REPORTING ONLY THE NAME OF THE MODEL

"GTR+I+G was used" is half the information. Missing are the criterion by which it was chosen, the number of gamma categories, and—if there were partitions—the scheme. The Block 12 report writes it out in full.

REMOVING CONSTANT SITES

They tell you nothing about the topology, but they do about the model. Removing them inflates branch lengths. The only situation in which they are removed is when the model knows it and corrects for it, which is what Mkv does in morphology.

BELIEVING THAT CHOOSING THE MODEL WELL FIXES THE TREE

It improves branch lengths, support and, in difficult cases, the topology. It does not fix a bad alignment, or a heterogeneous composition, or a taxon sampling that leaves a long branch on its own. The order of importance is: data, alignment, sampling, model.

4

CHAPTER FOUR · BLOCK 4

Parsimony and distances

The quick trees: the ones that come out in seconds and serve as a starting point for likelihood and as a contrast for everything else. A distance tree is not a weak hypothesis if the model that corrects the distances is the right one, and parsimony remains the right tool for morphology. This chapter explains both, their indices, how the support of a branch is measured and what a consensus says when there are many equally short trees.

4.1What quick trees are for

4.2Distance trees

4.3Maximum parsimony

4.4The search: NNI, SPR and TBR

4.5The indices: CI, RI, RC and HI

4.6Branch support

4.7Consensus and comparison

4.8The figure

4.9How well it does, measured

4.10Frequent mistakes

4.1 What quick trees are for

This block produces trees in seconds, and that makes them easy to look down on. It is not a good idea. They have three legitimate uses:

- **Starting point.** The maximum likelihood search of Block 5 needs an initial tree, and the better it is, the less risk there is of getting stuck in a local optimum.
- **Contrast.** If the distance tree, the parsimony tree and the likelihood tree agree on the main clades, that is information. If they disagree, so is that, and you need to understand why before publishing any of them.
- **The right tool for morphology.** With discrete characters and few changes, parsimony is not a cheap approximation to likelihood: it is the method the discipline uses, and with good reason.

> **A DISTANCE TREE IS NOT A WEAK HYPOTHESIS**
>
> It is when the distances are poorly corrected, which is a different matter. With the right model, neighbor joining is a **consistent estimator**: with enough data it converges to the true tree. Its bad reputation comes from using it with uncorrected distances, which is what does fail as soon as there is saturation.

4.2 Distance trees

The method consists of two decisions: **how the distance** between each pair of sequences **is measured** and **how the tree is built** from the matrix.

The distances

Table 4.1. The distances offered by the `distance` menu.

Distance	What it corrects and when to use it
uncorrected (p)	The proportion of differing positions. Useful only to see how much the others correct; not for inference.
Jukes–Cantor (JC69)	Multiple changes, assuming all changes are equally probable.
Kimura 2P (K80)	Separates transitions from transversions. The reasonable minimum for DNA.
Felsenstein 81	Allows unequal base frequencies.
Tamura–Nei 93	Unequal frequencies *and* two classes of transition. The most complete on the list.
LogDet (Lockhart)	The only one that **tolerates heterogeneous composition** among sequences. If the χ^2 test of Block 3 came out badly, this is the distance to use.
paralinear (Lake)	A close relative of LogDet, with the observed frequencies.
Poisson (proteins) and **Kimura (proteins)**	For proteins: the first without composition correction, the second with Kimura's classic approximation.

> **TWO NAMES FOR TWO DIFFERENT FORMULAS**
>
> What *ape* calls `logdet` is the distance of Lockhart et al. (1994); the formula with the observed frequencies is Lake's **paralinear** distance. They are often confused because they share the nickname. Here they appear separately and under their own names, so that you can report which one was used.

Two checkboxes modify the calculation. gamma correction applies the gamma correction with the α estimated by Block 3, and complete deletion decides what to do with gaps: when checked, it removes every column that has any gap before computing any pair; when unchecked, each pair uses the columns that both have. With a lot of missing data, complete deletion can leave you with very few columns.

The methods

The method menu offers five options: BIONJ, the one selected by default; neighbor joining (NJ), the classic method of Saitou and Nei; minimum evolution; UPGMA; and WPGMA, the weighted variant of UPGMA, which, when joining two groups, averages their distances giving them equal weight, regardless of how many sequences each one has.

> **! UPGMA AND WPGMA ASSUME A MOLECULAR CLOCK**
>
> And most data do not meet it. Both produce an **ultrametric** tree—all tips at the same height—, which is correct for a clustering dendrogram and false for a phylogeny with unequal rates. Use them to see structure, not to infer relationships. NJ and BIONJ do not assume a clock.

Balanced minimum evolution looks for the tree whose total length, computed with the balanced averages of Desper and Gascuel (2002) on Pauplin's (2000) criterion, is the smallest. It is a true optimality criterion, with NNI and SPR search from several starts, and not a mere clustering algorithm.

4.3 Maximum parsimony

The tree that needs the fewest changes to explain the matrix. Transparent, free of model assumptions… **and vulnerable for that very reason**, as you saw in Lab 1 of Chapter 1: without a model that corrects for multiple changes, parsimony interprets convergence as kinship.

The controls

- characters: unordered (Fitch), where any state can go to any other in one step; ordered (Sankoff), where the cost depends on the pair of states and an order can be imposed—useful in morphology, when a character has graded states—; or transversions weighted, which penalizes transversions more than transitions.
- tv:ts weight: how much more a transversion costs. With 2:1, a transversion is worth two steps. It is parsimony's way of acknowledging that transitions are more frequent.
- swapping: NNI, SPR or TBR (Section 4.4).
- starts: how many different addition orders are tried. Several starts that reach the same tree are the best sign that the search was sufficient.
- max trees: how many equally short trees are kept. Morphological matrices sometimes produce thousands.

4.4 The search: NNI, SPR and TBR

The number of topologies grows faster than factorially: with twenty taxa there are more than 10^{20} unrooted trees. They cannot all be evaluated, so the search starts from one and tries rearrangements.

Table 4.2. The three types of rearrangement, from smallest to largest reach.

Type	What it does	When
NNI	Swaps the two groups adjacent to an internal branch. Small moves.	Fast; easily gets stuck in local optima.
SPR	Pulls off a branch with everything hanging from it and regrafts it elsewhere.	The usual balance between reach and time.
TBR	Cuts the tree in two and rejoins the halves through every possible pair of branches.	The most exhaustive and the slowest.

No heuristic search **guarantees** the optimum. When two programs give different trees under the same criterion, it is usually this and not a disagreement of method: they are two local optima.

4.5 The indices: CI, RI, RC and HI

Once the tree is in place, four indices describe how much homoplasy—convergence and reversal—it contains.

> **THE FOUR DEFINITIONS**
>
> **CI** = m/s, the minimum possible number of changes over the number the tree needs. It equals 1 without homoplasy.
> **HI** = 1 − CI, the homoplasy index: the same thing said the other way round.
> **RI** = $(g–s)/(g–m)$, where g is the length on the worst possible tree: how much of the similarity that **could** be synapomorphy actually is.
> **RC** = CI × RI, the rescaled consistency index.

> **THE CI FALLS WITH THE NUMBER OF TAXA**
>
> Systematically, and without homoplasy having increased. A CI of 0.8 with fifteen taxa and one of 0.4 with two hundred may describe the same matrix. That is why **CIs are not compared between studies of different size**, and why the RI, which corrects part of the problem, is the one worth reporting.

4.6 Branch support

The sentence at the head of this section on screen deserves repeating: *a clade without support is not a bug: it is a polytomy your data cannot resolve*. The program offers three measures.

Bootstrap

The alignment columns are resampled **with replacement**, the tree is rebuilt hundreds of times and the percentage of replicates in which each branch appears is counted. It measures the **consistency of the signal** in the data, not the probability that the branch is correct. The convention: 70% for "well supported," 95% for strong.

Jackknife

The same, but instead of resampling with replacement a fraction of the columns is **deleted** in each replicate. It gives results similar to the bootstrap and is somewhat more conservative.

Bremer support

Different in nature: how many **extra steps** have to be accepted before finding a tree in which that branch no longer appears. It is read in steps, not as a percentage, and it depends on the size of the matrix: in a large matrix, a Bremer of 5 may be little. A value of 1 means that one step is enough to undo the branch.

> **BELIEVING THAT THE BOOTSTRAP PROTECTS AGAINST SYSTEMATIC ERROR**
>
> It does not, and it is important to understand this. The bootstrap measures whether the signal holds up when the same data are resampled under the same method. If the method is biased—long-branch attraction, heterogeneous composition—, the wrong signal is in **all** the replicates, and the bootstrap will support the wrong branch at 100%. With more data, it will support it more.

4.7 Consensus and comparison

When the search returns several equally short trees, the question is what they agree on. The menu offers four consensus methods:

Table 4.3. The four consensus methods.

Consensus	Keeps
strict	Only the clades present in **all** trees. The most conservative and the one published by default.
majority rule (50%)	The clades present in more than half, with their frequency noted.
semistrict	The clades that no tree contradicts, even if they are not in all of them. Useful when some trees lack taxa.
greedy	Adds clades by frequency as long as they are compatible. The most resolved and the least conservative.

The polytomies of a strict consensus **are the result**: they say exactly where the data do not decide. A heavily collapsed consensus is not a failure of the analysis; it is an honest analysis of data that are not enough.

The **How much the trees agree** table compares by Robinson–Foulds distance everything the block has produced—distances, parsimony, consensus trees—to see at a glance which methods agree.

4.8 The figure

The drawing uses the same engine that the tree studio of Block 12 will use: rectangular, circular or unrooted layout; cladogram to draw without lengths; support as numbers, colored dots or none; and root as it is, midpoint or outgroup…, with multiple selection using Ctrl.

> **AN UNROOTED TREE HAS NO ANCESTORS**
>
> The methods in this block produce **unrooted** trees: they describe relationships, not direction. Placing the root is an additional decision, and with an outgroup it is a hypothesis that can be wrong. If you choose an outgroup that is not monophyletic in that tree, the program says so and leaves the tree as it was, instead of silently rooting it wrongly.

4.9 How well it does, measured

Everything in this block was checked against *ape* and *phangorn*, and the results are in the validation folder. Three deserve mention:

- The parsimony search reaches **1,916 steps** on the example, **the same optimum as phangorn's** `pratchet`, and thirteen steps better than `optim.parsimony` with SPR.
- The balanced minimum evolution length reproduces that of `ape::fastme.bal` **to within $5{\cdot}10^{-9}$** on the same tree, so the formula agrees exactly.
- With its own search from three starts, the program reaches a **shorter** tree (2.899076) than ape's (2.901969). It is not that one of them is wrong: they are two local optima of the same criterion, and where a heuristic lands depends on where it starts.

4.10 Frequent mistakes

> **USING UNCORRECTED DISTANCES**
>
> It is what gave distance methods their bad reputation. With saturation, the p distance underestimates divergence exactly where it matters most, and the tree gets distorted. Always correct, and with the model that Block 3 chose.

✕ PUBLISHING A UPGMA TREE AS A PHYLOGENY

UPGMA assumes a molecular clock. If your rates are unequal—and they almost always are—, the tree will be wrong in a systematic and predictable way: the fast lineages will come out more "derived" than they are.

✕ REPORTING A SINGLE SHORTEST TREE WITHOUT SAYING HOW MANY THERE WERE

If the search found 4,800 equally short trees and you publish one, you are presenting as a result an arbitrary choice among 4,800. What is reported is the consensus, the number of optimal trees and their length.

✕ COMPARING THE CI WITH THAT OF ANOTHER STUDY

It is only comparable between matrices of the same size. Report the RI, and in any case say how many taxa and how many characters the matrix had.

✕ ACCEPTING THE SEARCH WITH A SINGLE START

A single starting point and NNI is the recipe for getting stuck in a local optimum. Raise the starts, use SPR or TBR, and check that several starts reach the same length. If they do not, the search is not over.

5

CHAPTER FIVE · BLOCK 5

Maximum likelihood

The tree that gets published. Under an explicit model, likelihood searches for the topology and branch lengths that make the observed data most probable, and—unlike parsimony—it is **consistent** as long as the model is reasonable. This chapter explains the search, the fit with one model per partition, the **four** support measures with their four different thresholds, and how to formally test another author's hypothesis.

5.1What it optimizes, and why it matters

5.2The model and the starting tree

5.3The search: NNI, SPR and the radius

5.4Reading the result

5.5One model per partition

5.6The four support measures

5.7Testing a hypothesis

5.8Threads, and why it takes as long as it does

5.9How well it does, measured

5.10Common mistakes

5.1 What it optimizes, and why it matters

The likelihood of a tree is the probability of having observed **these data** if that tree and that model were true. The maximum-likelihood tree is the one that makes that probability highest. It is reported as a logarithm, always negative, and less negative is better.

> **A LNL ON ITS OWN MEANS NOTHING**
>
> Its absolute value depends on the number of sites and taxa. It only makes sense to **compare** likelihoods computed on exactly the same data. A lnL of −18,674 is neither "bad" nor "good": it is the one for these data under this model, and it is only useful against another from the same alignment.

The decisive advantage over parsimony is the one from Chapter 1: under the correct model, likelihood is **consistent**, that is, with enough data it converges to the true tree even when there are unequal long branches. Parsimony, in that same geometry, converges with growing certainty to the wrong tree.

The condition "under the correct model" is not rhetoric. If the model is misspecified—unmodeled compositional heterogeneity, uncorrected saturation—likelihood is wrong too, and with more data it is wrong with more confidence. That is why Block 3 comes before this one.

5.2 The model and the starting tree

The first row of controls sets the model: model (GTR, HKY, LG…), +G with so many categories, +I, and the frequencies, which can be empirical—counted in the alignment—or equal. The Use the model chosen in Block 3 button copies what won there, which is almost always the right thing to do.

> **THE ESTIMATE THEM (+F) CHECKBOX**
>
> By default the base frequencies are **counted** from the alignment, which is what `phangorn::optim.pml` does without `optBf`, and that is why the program's results match its own. Turning on **+F** estimates them by maximum likelihood together with everything else, which usually gives a slightly better lnL. It is off by default to preserve that comparability, not because it is worse.

The starting tree can be the parsimony (Block 4) tree, the distance (Block 4) tree, BIONJ, or a random one. This is no minor detail: the search is heuristic, and where it starts determines which local optimum it may get stuck in. Trying two different starting points and reaching the same lnL is the best evidence that the search was enough.

5.3 The search: NNI, SPR and the radius

The search starts from the starting tree and tries rearrangements: first **NNI**, which swaps neighboring branches, and then **SPR**, which cuts a whole branch off and grafts it back somewhere else. The rearrangements menu lets you use NNI only or NNI + SPR.

Two controls govern the cost:

- SPR radius: the maximum distance, in number of branches, at which regrafting is allowed. A small radius is fast and explores little; a large one explores a lot and takes long. Five is a reasonable balance.
- max rounds: how many complete passes of rearrangements are made before stopping.

> ∑ WHY THE SEARCH IS FAST WITHOUT CHEATING
>
> Scoring each candidate by reoptimizing the whole tree would be unfeasible. The program does what PhyML and RAxML do: for each move, it **reoptimizes only the branches the move touched**, relying on the incremental computation of the likelihood engine. SPR also runs in **two passes**: first all candidates are scored with the inherited branch lengths—RAxML's "lazy SPR"—and then only the best dozen are scored again with their neighborhood reoptimized. The winning tree is refitted in full at the end.

5.4 Reading the result

The tiles summarize the run: the **log-likelihood**, the number of **parameters** (branches plus model), **AIC** and **BIC**, how many **rounds** were needed and how many topologies were evaluated, the **time**, and the estimated model parameters—the gamma **α** and **κ** or the relative rates.

Below the tiles, the panel writes out in prose what they mean. Two things deserve attention:

- The number of **rounds**. If the search ended in a single round, it may not have explored anything; if it used up the maximum, it was probably still improving and you should raise it.
- The **α**. Below 1 there is strong heterogeneity among sites, and not modeling it would have distorted the long branches.

5.5 One model per partition

A chloroplast gene and nuclear ITS do not evolve at the same rate nor with the same composition. The partitioned fit acknowledges this without breaking the tree: all partitions **share the topology and the branch proportions**, but each has its own model and a **rate multiplier**, normalized to a site-weighted mean of one.

So a partition with a multiplier of 0.3 evolves at a third of the mean rate, and one with 2.5, two and a half times faster.

WHY THE FIT STARTS OUT "TIED"

Releasing all partitions from the very start made the search fall into a worse optimum. The program starts with all of them sharing one model during the first passes and releases them afterward. It is an implementation detail, but it explains why the partitioned result never falls below the single model it contains.

PARTITIONING DOES NOT RESOLVE A REAL CONFLICT

Giving each gene its own rate does not free it from **sharing the tree**. If two genes have genuinely different histories—chloroplast capture, hybridization—the partitioned analysis averages them more elegantly, but it averages them all the same. Uncovering them is the job of Block 11.

5.6 The four support measures

This is the section of the chapter that prevents the most mistakes. The program computes four measures, and **they do not mean the same thing nor are they read against the same threshold.**

Table 5.1. The four support measures, with their conventional threshold.

Measure	Threshold	What it measures
Classical bootstrap	**≥ 70**	In what percentage of alignments resampled with replacement the branch reappears. Conservative: 70 is already fair support.
Ultrafast bootstrap (UFBoot)	**≥ 95**	An approximation by RELL, built to be **nearly unbiased**. Its scale is not that of the classical bootstrap.
SH-aLRT	**≥ 80**	Compares the likelihood of the tree with that of the two NNI rearrangements of that branch. A different question, not another version of the same one.
aBayes	—	A fast Bayesian approximation on the same local comparison.

THE MOST COMMON MISTAKE WITH UFBOOT

Reading it with the classical bootstrap threshold. Because UFBoot is nearly unbiased, it is **less conservative**: a UFBoot of 90% is not equivalent to a bootstrap of 90% but to considerably **less** support. That is why its threshold is 95 and not 70. The header of the on-screen table has the threshold written in, precisely so they are not confused.

Current practice in the literature is to require **both**: SH-aLRT ≥ 80 **and** UFBoot ≥ 95. They are different information—one measures whether the branch improves the likelihood, the other whether the signal holds up under resampling—and a branch that passes both is much better supported than one that passes only one.

> **WHAT NONE OF THE FOUR MEASURES**
>
> Whether the tree is correct. All four measure the **internal consistency** of the signal under the method and model used. If the model is wrong, all four will support the wrong branch, and with more data they will support it more. High support is necessary, not sufficient.

5.7 Testing a hypothesis

Here the block stops describing and starts testing. The question is: do my data **reject** the tree proposed by another author, or the one my hypothesis proposes?

It is done in two steps:

1. Select the taxa of the group being tested and press Search for the best tree under that constraint: the program searches for the best tree **forcing** that group to be monophyletic. That is the best possible case for the hypothesis.
2. Press Compare the trees (KH, SH, AU): the three tests contrast the constrained tree against the unconstrained one, using RELL replicates.

A p-value below 0.05 rejects the alternative tree. Of the three, the **AU** test is the one that best corrects for selection bias and the one you should report; KH and SH are there because they are still requested.

> **"NOT REJECTED" IS NOT "JUST AS GOOD"**
>
> With few data almost nothing is rejected, and with many almost everything is. A nonsignificant result says that **your data are not enough to rule out** that hypothesis, not that it is correct. And the set of trees to compare must be fixed **before** looking at the results: choosing afterward what to compare is exactly the bias the AU test tries to correct.

5.8 Threads, and why it takes as long as it does

The bootstrap and the Bayesian chains are spread over browser **threads**, built from the program's own code. Two measured facts are worth knowing:

- A freshly created thread **runs interpreted** and only reaches optimized code after a few seconds of repeating the same loops. That is why **16 replicates on 16 threads take longer than the same 16 on six**: each thread pays its whole warm-up and throws it away. The program never assigns fewer than three jobs per thread.
- One hundred maximum-likelihood bootstrap replicates take minutes. This is JavaScript, not C, and the page says so before starting, with an estimate and a cancel button.

If the browser does not allow threads—some configurations block them—the work is done in the page, slower but without failing.

5.9 How well it does, measured

The validation of this block checks against *phangorn* on the same rbcL alignment. The result, which is in the validation folder and can be reproduced:

Table 5.2. The maximum-likelihood search compared with phangorn 2.12.1, on rbcL.

Search	lnL reached	Comment
`phangorn::optim.pml` with NNI	−7126.989550	The package's standard search.
`phangorn`, stochastic search	−7108.802104	Its most exhaustive search.
PhylogenyPro, NNI + SPR, 52 s	**−7108.800752**	Better than NNI by 18 units, and slightly better than the stochastic one.

The difference from the NNI search—eighteen log-likelihood units—is not a nuance: it is the difference between staying in a local optimum and getting out of it, and it is exactly what SPR buys.

5.10 Common mistakes

READING UFBOOT WITH THE BOOTSTRAP THRESHOLD

Already said, and repeated because it is the most common: 70 for the classical bootstrap, **95** for the ultrafast one, 80 for SH-aLRT. Three measures, three thresholds.

SEARCHING ONLY ONCE, FROM A SINGLE STARTING TREE

The search is heuristic and does not guarantee the optimum. If two different starting points reach different lnL values, the higher one rules and the search had not finished. Reaching the same one is the only reasonable evidence that it had.

COMPARING LNL ACROSS DIFFERENT ALIGNMENTS

Removing three columns changes the lnL, and not because the tree is worse. Likelihoods are only compared on exactly the same data: same alignment, same columns, same taxa.

PRESENTING A CONSTRAINED TREE AS "THE TREE"

The constrained tree is the best case **for the hypothesis being tested**, not an estimate. It is computed in order to compare it, and what gets published is the comparison—the lnL of both and the p-value of the AU test—not the constrained tree by itself.

BELIEVING THAT HIGH SUPPORT VALIDATES THE MODEL

It is the other way around: support is computed **assuming** the model. If Block 3 warned of compositional heterogeneity or saturation, a UFBoot of 100 does not refute it; it repeats it with more confidence.

6

CHAPTER SIX · BLOCK 6

Bayesian inference

Not “the best tree,” but the probability of each tree given the data **and what was assumed before seeing them**. That second term is what sets the method apart and what forces you to write down the assumptions that other methods leave implicit. This chapter explains the priors, the chain and its coupling, the four numbers that tell whether it converged, how posterior probabilities are read—they are not bootstrap values—and Bayes factors.

6.1What changes with respect to likelihood

6.2What is assumed before seeing the data

6.3The chain and its coupling

6.4Did it converge? The four numbers

6.5Posterior probabilities

6.6Bayes factors

6.7Taking the result away

6.8How the sampler was checked to be correct

6.9Common mistakes

6.1 What changes with respect to likelihood

Maximum likelihood asks: *which tree makes these data most probable?* Bayesian inference asks the opposite: *how probable is each tree, given these data?* To answer that, something more is needed: a **prior distribution** stating what was believed before looking at the data.

The result is not a tree but a **distribution** over trees, branch lengths and parameters. From it come the posterior probabilities of each clade and the credible intervals of each age, which is what makes this block the natural companion of Block 7.

> ✦ THE PRIOR IS NOT A WEAKNESS OF THE METHOD: IT IS AN OBLIGATION TO WRITE IT DOWN
>
> Those who criticize the Bayesian approach for "depending on assumptions" tend to forget that likelihood also makes assumptions—the model, independence among sites, compositional homogeneity—it just does not force you to declare them. Here you have to write them down, and that is an advantage: the assumptions are in plain sight and can be discussed.

6.2 What is assumed before seeing the data

The first section of the panel is the list of priors. The `Use the model from Block 5` button copies the model already chosen; the next three fields set the priors:

- `mean branch length (exponential)`: the mean of the exponential on each branch length.
- `mean of α (exponential)`: the prior on the shape of the gamma.
- `mean of the rates (exponential)`: the prior on the rates of the substitution model.

In addition, the program uses a **uniform prior on topologies**—all equally probable before seeing the data—and a **Dirichlet** on the base frequencies.

> ! THE BRANCH-LENGTH PRIOR IS NOT HARMLESS
>
> It is the assumption that most quietly bends a result. With the exponential of mean 0.1—MrBayes's default—and **divergent** data, the posterior of the total tree length comes out at **half** that of the maximum-likelihood tree: the prior pulls the branches down and wins. This is documented (Brown et al. 2010; Marshall 2010), and the panel warns about it before running, comparing the prior with the length that Block 5 measured.
>
> The solution is not to remove the prior—that cannot be done—but to **choose it deliberately** and check that the result does not change with another reasonable one.

6.3 The chain and its coupling

The sampler is Metropolis–Hastings over topologies, branch lengths and model parameters. At each step it proposes a change and accepts or rejects it according to a rule that guarantees that, in the long run, the frequency with which it visits each tree is that tree's posterior probability.

There are six controls:

Table 6.1. The chain controls.

Control	What it is
generations	How many steps each chain takes. More is better and slower.
burn-in	The first generations, while the chain climbs from where it was released, are thrown away.
sample every	How many steps between saved samples. Saving all of them adds no information, because they are correlated.
runs	How many **independent** chains, started separately. With fewer than two, convergence cannot be checked.
chains per run	The coupled chains (MC³) of each run.
heating	How much the posterior of the heated chains is "flattened."

> **∑ WHAT COUPLING (MC³) IS FOR**
>
> A single chain can get trapped on one peak of the posterior and never leave, because to reach another peak it would have to cross a valley of low probability. Coupling runs several chains at once: one **cold** chain, which is the one that samples, and several **heated** ones that see a flattened posterior and move much more freely. Every so often, two chains swap positions. This way the cold chain can jump to a peak the heated one found, without ceasing to sample the correct distribution.

Each run executes in its own browser thread, so four runs make use of four cores.

6.4 Did it converge? The four numbers

The program puts the diagnostics **before** the tree, and not on a whim: a Bayesian tree from a chain that did not converge is a pretty picture of nothing.

Table 6.2. The four convergence numbers and their threshold.

Number	Target	What it says
ESS	**≥ 200**	How many **independent** samples the ones the chain took are worth. Ten thousand correlated samples may be worth thirty.
PSRF ($\hat{R}$)	**< 1.01**	Compares the variance within each chain with the variance among independent chains. If all explore the same thing, it tends to 1.

Table 6.2. The four convergence numbers and their threshold. *(continued)*

Number	Target	What it says
ASDSF	**< 0.01**	How much two runs contradict each other on the support of each branch. It is the number MrBayes prints while running.
The trace	—	The plot of the log-likelihood against generation. Where it stops climbing and starts to oscillate, that is where burn-in ends.

> ✕ TRUSTING A FLAT TRACE
>
> A flat trace does not prove convergence: it may be a **stuck** chain that has spent a million generations without leaving a local optimum. That is why you need **several independent runs** and the PSRF, which compares among them. And the PSRF needs the runs to start at **different** places: if all of them start from the same tree, they can give 1.00 and all be equally wrong.

A high ESS is not enough on its own either: it says that the chain **mixed well where it was**, not that it covered the whole space.

6.5 Posterior probabilities

The tree that is reported is the **majority-rule consensus** of the samples, with the frequency of each clade as its posterior probability. The conventional support threshold is **0.95**.

> ! A POSTERIOR IS NOT A BOOTSTRAP
>
> They are different scales and are not compared. For the same branch, the posterior probability is usually **considerably higher** than the bootstrap: it is not that the Bayesian method "supports more," it is that it answers another question. Moreover, the posterior is known to be **overconfident** when the model is misspecified: it can give 1.00 to a branch that the bootstrap supports at 60%. A 1.00 is not a guarantee; it is a statement conditional on the model and the priors.

The panel also shows the **credible set** of trees: how many distinct trees are needed to accumulate 95% of the posterior. If three are needed, the result is firm; if eight hundred are needed, the topology is far from resolved, however well supported the consensus may look on some branches.

6.6 Bayes factors

AIC and BIC compare models with a shortcut: they penalize the number of parameters. The Bayes factor compares them by **integrating over all possible values of those parameters** according to the prior, which is more costly and more correct.

The quantity being integrated is the **marginal likelihood**, and the program estimates it by **stepping stones** (Xie et al. 2011): a series of intermediate distributions between the prior and the posterior, with the steps and generations per step controls. Never by the harmonic mean, which is notoriously bad.

Table 6.3. The Kass and Raftery scale on 2·ln(BF).

2·ln(BF)	Evidence in favor of the better model
less than 2	Not worth mentioning.
2 to 6	Positive.
6 to 10	Strong.
more than 10	Very strong.

> **COMPARING BAYES FACTORS COMPUTED WITH DIFFERENT PRIORS**
>
> The marginal likelihood is **much more sensitive to the prior** than the posterior, because it integrates over it. Two people with the same model and different priors can obtain opposite Bayes factors. And the estimate has its own error: repeating it with another seed tells you how much, and it is worth doing before publishing a 2·ln(BF) of 7.

6.7 Taking the result away

The block exports the consensus tree with its posterior probabilities and a **Tracer-compatible log**, so the diagnostics can be reviewed with the tool the community uses. The tree passes on to Block 7 just like the maximum-likelihood one, and for dating it is often preferable, because the Bayesian relaxed clock gives a distribution per node instead of a point.

6.8 How the sampler was checked to be correct

An MCMC sampler is hard to validate: there is no reference number to compare against, because the result is a distribution. This block was checked in three different ways, and all three are in the validation folder.

1 · Switching off the likelihood

If the likelihood is disconnected, the sampler **must reproduce its own prior exactly**, because there are no longer any data to pull it away. It is the most severe test it can be given, and it passes. In the recorded run—400,000 generations; the bench is stochastic, so another run gives different last decimals—:

- The **three unrooted topologies** of four taxa come out at 0.3374, 0.3335 and 0.3292, against an expected 1/3.
- With six taxa, **all 105** possible topologies are visited, with a mean frequency of 0.00952, against 1/105 = 0.00952.
- Branch lengths give a mean of 0.10060 against the exponential of mean 0.1, and the total tree length 0.50302 against the 0.5 of the corresponding Gamma(5, 0.1).

2 · Comparing the diagnostics with *coda*

The HPD and PSRF are computed on a series that R and the browser build in the same way, and they come out **identical** to those of `coda`, which is the reference implementation.

3 · Auditing the chain move by move

An audit compares the likelihood the chain keeps on record—computed incrementally, for speed—with a computation from scratch, at every move. The mismatch is **zero** in all of them.

> **A FINDING ONLY THE SAMPLER COULD UNCOVER**
>
> During those tests the chain gave likelihoods **thirty units above** the supposed maximum of Block 5. It was not a bug: the chain **estimates the base frequencies**, while the maximum-likelihood fit kept them fixed at the empirical ones. That is where the **+F** checkbox of Block 5 came from, which now lets you estimate them there too.

6

6.9 Common mistakes

> **LOOKING AT THE TREE BEFORE THE DIAGNOSTICS**
>
> The panel puts them in that order on purpose. With an ESS below 200, a PSRF above 1.01 or an ASDSF above 0.01, you do not look at the tree: you lengthen the run.

> **RUNNING A SINGLE CHAIN**
>
> Without at least two independent runs there is no PSRF or ASDSF, and therefore no way of knowing whether it converged. A single chain with a beautiful trace may have been stuck for a million generations.

> **REPORTING A POSTERIOR AS IF IT WERE A BOOTSTRAP**
>
> "95% support" without saying which measure it is invites the reader to read it on the wrong scale. Write "posterior probability of 0.95," and if there is also a bootstrap, give the two separately.

LEAVING THE DEFAULT BRANCH-LENGTH PRIOR WITHOUT LOOKING AT IT

With divergent data, the exponential of mean 0.1 can halve the tree length. The panel compares the prior with what Block 5 measured and warns **before** running: that warning must be read.

DISCARDING 25% OUT OF HABIT

Discarding a quarter of the generations is the convention, but **looking at the trace is better than the convention**. Sometimes burn-in ends at 2% and good data are being thrown away; sometimes it has not ended at 50%.

7

CHAPTER SEVEN · BLOCK 7

The molecular clock and dating

A phylogram measures change; a chronogram measures time. Getting from one to the other needs three things that molecular data do **not** contain: a root, a model of how the rate varies, and at least one external calibration. This chapter goes through those three steps in order, explains the assistant that reviews calibrations before they are used, and ends by insisting on the one thing that must never be missing from a published date: its interval.

7.1What the clock cannot give you on its own

7.2The root

7.3Does the clock hold?

7.4The calibrations

7.5The calibration assistant

7.6The three dating methods

7.7Reading the chronogram

7.8How well it does, measured

7.9Common mistakes

7.1 What the clock cannot give you on its own

The branches of the tree produced by Blocks 5 and 6 are measured in **substitutions per site**. Not in years. A branch of 0.08 can be a million years of fast evolution or fifty of slow evolution, and the sequences, by themselves, **cannot tell** those two cases apart.

> **THE CONFOUNDING OF RATE AND TIME**
>
> What a branch measures is **rate × time**. With one observed number and two unknowns, outside information is needed to separate them. That is the whole difficulty of molecular dating, and it explains why a date depends more on its calibrations than on any statistical refinement.

That is why the block asks for three things in this order: a **root**, because without it there is no direction and no ages; an answer to **does the clock hold?**, which decides which method to use; and at least one **calibration**, without which ages come out as proportions of the depth of the tree and not in years.

7.2 The root

A tree without a root has no direction and therefore no ages. The root by menu offers four routes:

Table 7.1. The four ways of placing the root.

Method	When, and with what care
outgroup	The honest way when you have one. It is a hypothesis, and it can be wrong: an outgroup that is too distant introduces a long branch and can attract the most divergent taxon of the ingroup.
midpoint	Places the root at the center of the longest path. Convenient, but it **assumes a clock**—exactly what is about to be tested. Use it to explore, not to publish.
tip dates	When the tips have different dates—viruses, ancient DNA, dated fossils—the data themselves say where the root goes. You write one date per taxon, one per line.
leave it as it is	If the tree already came rooted from another block or from a file.

The root-to-tip regression

With dates at the tips, the program plots the distance from the root to each tip against its date. If there is a clock, the two rise together in a straight line and **the slope is the rate**. The r^2 tells how much of the variation time explains.

> **THE R^2 OF THIS REGRESSION IS OPTIMISTIC**
>
> The root-to-tip distances of two sister taxa are **not independent**: they share almost the whole path. That is why the r^2 comes out higher than it should and **must not be used as a formal test** of the clock. It is useful for looking, for spotting odd tips and for estimating an approximate rate.

7.3 Does the clock hold?

The test compares two fits of the **same tree**: one with free branches, one length per branch, and another forced to be **ultrametric**, where all tips sit at the same height. The difference in log-likelihood, multiplied by two, is tested against a χ^2 with $n-2$ degrees of freedom.

A small p-value rejects the strict clock and calls for a relaxed clock.

> **READING "NOT REJECTED" AS "THERE IS A CLOCK"**
>
> It does not prove it. The test has **little power with few taxa**: with twelve species it rejects almost nothing. Conversely, with many sites it almost always rejects, because it detects tiny deviations of no practical importance. The panel says so in those words when the result is not significant.

7.4 The calibrations

Without at least one, ages do not exist: substitutions per site and millions of years are two different scales and nothing links them. The program accepts four sources of years and, in the type menu, five prior shapes.

Table 7.2. The five prior shapes and what they assert.

Shape	What it says about the node
fixed age	It has exactly this age, with no uncertainty. Almost never true in biology.
uniform interval	It lies between these two ages, with no preferred point. The shape of a geological event.
normal	It lies around a mean, with a deviation (mean / M and sd / S), equally likely toward younger and older ages. The shape of an age taken from another study; never that of a fossil, because it would allow the node to be younger than the fossil.
offset lognormal	It is no younger than the offset, with a long tail backward. The shape of a fossil.
offset exponential	The same, with the mass more concentrated near the offset.

> **A FOSSIL GIVES THE *MINIMUM* AGE, NOT THE AGE**
>
> This is the idea to take away from this chapter. When that individual died, the lineage **already existed**, and it may have existed long before: what the fossil proves is that the node is no younger than it. That is why asymmetric, offset priors are used, which allow the node to be much older but not younger. Entering the fossil's age as a **fixed age** asserts that the lineage arose exactly when that individual died, and squeezes the whole tree toward the present.

To select the node you choose the taxa of the clade —none means the root. If those taxa do not form a clade in the tree, the program says so instead of calibrating something other than what was asked.

7.5 The calibration assistant

Above the calibration controls there is an assistant whose purpose is that nobody has to guess what M and S mean. **It decides nothing**: it tells you what you have just written.

It proposes the shape according to where the age comes from

You choose it in the age comes from and the program explains why that source calls for that shape:

Table 7.3. What the assistant proposes according to the source.

The age comes from	It proposes	Because
A fossil assigned to this clade	offset lognormal	It is a minimum bound, not a point.
A geological event	uniform interval	It is a window with a beginning and an end.
An age taken from another study	normal	It carries the uncertainty of that work, and it is wise to **widen** it, not narrow it.
An age known for certain	fixed age	Only to fix a reference scale.

It says, in years, what the prior asserts

This is the most useful part. "Offset lognormal, M = 1.6, S = 0.7, offset 20" means nothing to almost anyone; below the controls the translation appears, recalculated with every keystroke:

→ WHAT YOU READ ON SCREEN

*This prior says the node's age lies, with 95 % probability, between **21.3** and **39.5** Ma, with a median of **25.0** Ma, and cannot be younger than 20.0 Ma.*

And it warns about the mistakes that ruin a dating

- A **fossil entered as a fixed age**, or with a symmetric normal prior, which would allow a node younger than the fossil.
- An **offset of zero**, which imposes no minimum age: it allows the node to be from yesterday.
- A **normal whose lower tail reaches into the future**, that is, into negative ages.
- A prior **so wide it barely informs**, or so narrow it feigns a certainty nobody has.
- The confusion of **crown and stem**: the taxa you select calibrate the crown node—the most recent common ancestor of the group. If your fossil is **sister** to the group and not a member of it, what it calibrates is the **stem**, which is older, and for that you also have to select the sister group.
- That this is going to be the **only** calibration, and therefore that the whole scale of the tree depends on it.

WHY THERE IS NO FOSSIL DATABASE

And there will not be one. Assigning a fossil to a node is a taxonomic judgment about that fossil's characters and about which clade it is sister to: it is the part that is debated in papers and the part a reviewer examines. A lookup table would invite publishing a date that nobody looked at. The program helps with everything surrounding that judgment; the judgment is yours.

7.6 The three dating methods

Three routes, from the fastest to the most honest about uncertainty.

Least squares

Fits the ages by minimizing a least-squares function on the branch lengths (To et al. 2016). It gives a point estimate in seconds, even with five hundred taxa. It is what you use to see the order of magnitude, or when the tree is too large for the others.

Penalized likelihood

Lets the rate change between branches but **penalizes** abrupt changes (Sanderson 2002). The smoothing λ parameter controls how much: a large λ approaches a strict clock, a small λ lets each branch have its own rate.

λ IS NOT CHOSEN BY EYE

It is chosen by **cross-validation**, removing one branch at a time and seeing which λ predicts best. The choose it by cross-validation button does exactly that and plots the curve. If the minimum of that curve falls at one end of the range tried, the range was too narrow and has to be widened.

Bayesian relaxed clock

An uncorrelated lognormal clock (Drummond et al. 2006): each branch draws its rate from a lognormal, independently of its neighbor. It is the slowest and the only one that returns a **distribution per node** instead of a point, which is what should be published when calibrations are uncertain—that is, almost always.

7.7 Reading the chronogram

The tiles summarize: the **root age**, the **method**, the computing **time**, the **mean rate** in substitutions per site per unit of time, and the **rate variation**, a coefficient of variation that says how far the tree departs from a strict clock: near 0, there practically was a clock; above 0.5, rates are very unequal and a strict clock would have given wrong dates.

The chronogram is drawn on the **geological time scale**—the bands of the International Chronostratigraphic Chart—with uncertainty bars at each node.

> **PUBLISHING AN AGE WITHOUT ITS INTERVAL**
>
> A node dated at 30 Ma with an interval of 28 to 32 is well determined; the same 30 Ma with an interval of 8 to 70 says almost nothing, even though the mean looks just as round. **An age without an interval is not a result.** And the interval only captures the uncertainty *within* the model and the calibrations you gave: it does not include the error of having chosen a calibration badly, which is usually the largest of all.

Without any calibration, the panel fixes the root at 1 and warns that the ages are **proportions of the depth of the tree**, not years: they serve to compare nodes with each other, not to write "X million years ago."

7.8 How well it does, measured

Checked against *ape* and *phangorn*, with the results in the validation folder:

- **Rooting by tip dates reproduces** `ape::rtt`: slope −0.00158106 against −0.00158106, with a difference of $3{\cdot}10^{-9}$, and r^2 0.783919 against 0.783919.
- **Penalized likelihood agrees with** `ape::chronos`: over 15 clades, the maximum age difference is 0.0382 with the root at 1, and the correlation between the two series of ages is $r^2 = 0.9906$.
- In the clock test, the program's own ultrametric fit is **better than that of** `optim.pml(optRooted = TRUE)` **by 11.8 units** of log-likelihood on the same rooted tree, and both reject the clock.

> **A BUG THE VALIDATION UNCOVERED**
>
> The rooting search filtered by the **sign** of the slope, assuming that root-to-tip distance grows with the date. It grows with calendar dates and **shrinks** with ages counted backward, so the filter discarded all the correct rootings: it gave an r^2 of 0.006 where ape reaches 0.784. Now the sign only indicates which convention the dates follow, and the rate is the absolute value of the slope.

7.9 Common mistakes

> **USING THE MIDPOINT AND THEN TESTING THE CLOCK**
>
> Midpoint rooting **assumes** a clock. Testing the clock on a tree rooted that way is asking the method what the method already took for granted. With an outgroup, the test is legitimate.

> **A SINGLE CALIBRATION**
>
> The whole scale of the tree hangs on it, and its error propagates to every node with nothing to offset it. Two or three well-spread calibrations are much better than one, however good, because they also check each other.

CALIBRATING WITH A SECONDARY AGE WITHOUT WIDENING ITS UNCERTAINTY

An age taken from another study carries the tree, the model and the calibrations of that work. Copying its interval as it is pretends that uncertainty does not exist. The assistant proposes a 12% deviation and recommends widening it, not reducing it.

CONFUSING CROWN AGE WITH STEM AGE

The crown is the most recent common ancestor of the living species of the group; the stem, its divergence from the sister group, and it is always older. The diversification rates of Block 8 are computed with one or the other and **give different results**. You have to say which one was used.

REPORTING ANOTHER STUDY'S RATE AS IF IT WERE DATA

A rate published for "the same marker" was estimated in another group, with other sampling and other calibrations. It is the largest source of error in a dating and the least reported.

8

CHAPTER EIGHT · BLOCK 8

Diversification

A chronogram does not only say when: the way its branching times are distributed says how fast the group diversified, and whether that pace changed. This chapter explains the lineages-through-time curve, the γ statistic and the trap that comes with it—**missing species imitate a slowdown exactly**—the five models that are compared, the estimators that work with a single age and a richness, and why extinction rates from a tree of living species cannot be trusted.

8.1What is read in the branching times

8.2Lineages through time and the γ statistic

8.3The incomplete-sampling trap

8.4The five models

8.5λ, μ, r and ε: what can be believed

8.6When there is only an age and a richness

8.7A rate for every species: DR

8.8How well it does, measured

8.9Common mistakes

8.1 What is read in the branching times

This whole block is read off the **branching times** of the chronogram, so you need to have gone through Block 7. Without dates there is nothing to analyze: lengths in substitutions do not say at what pace diversity appeared.

The question being answered is a single one, in many forms: did the group always diversify at the same pace, or was there an early burst that slowed down, or a recent acceleration? And behind it, the uncomfortable one: how much extinction was there, and can it be known?

> **! THE TREE BEING ANALYZED IS THE SURVIVORS' TREE**
>
> A chronogram of living species is a **reconstructed tree**: the lineages that went extinct left no tips and are not there. Almost all the difficulty of the block comes from that amputation, and it explains why net diversification is estimated well and extinction, badly.

8.2 Lineages through time and the γ statistic

The **lineages-through-time** curve counts how many lineages existed at each moment in the past. Drawn on a **logarithmic scale**, a constant rate gives a **straight line**: that is the reference against which everything else is judged.

- If the curve **bends down** near the present, the nodes piled up early: the group diversified early and then slowed down.
- If it bends up, the opposite happened.

Pybus and Harvey's (2000) **γ** statistic puts a number on that. It compares where the nodes fall with where they would fall under a constant rate. Under a constant rate γ follows a standard normal distribution, so that:

> **∑ HOW Γ IS READ**
>
> **γ < −1.645** rejects a constant rate in favor of a **slowdown**, one-tailed, at 5%.
> Between −1.645 and 1.645, compatible with a constant rate.
> Above 1.645, acceleration toward the present.

If there are several trees—from a bootstrap or from a posterior—the panel also draws a **band** over the sample, which is much more informative than the curve of a single tree.

8.3 The incomplete-sampling trap

This is the section that saves papers.

> **✦ MISSING SPECIES IMITATE A SLOWDOWN *EXACTLY***
>
> When the whole group is not sampled, the missing species come mostly **from the recent nodes**, because young clades have few species and losing one is enough to erase a node. The result is a curve that flattens near the present—indistinguishable from a real slowdown. With 60% sampling, γ will come out negative even if the rate was perfectly constant.

That is why the **MCCR test** (Monte Carlo constant rates) exists. Instead of comparing γ against the standard normal, it simulates constant-rate trees **with the same incomplete sampling as the data**, measures their γ, and compares the observed γ against that distribution. The panel asks for:

- species in the clade (in total, not those in the tree): the real number of described species, which is the figure you have to look up in the literature.
- replicates: how many trees are simulated to build the null distribution.

The result is a **critical value at 5%** that will be **more negative** than −1.645, and how much more negative it is says how much slowdown the sampling explained on its own. If your observed γ does not cross that critical value, there is no slowdown to report.

> ✕ GIVING THE WRONG TOTAL NUMBER OF SPECIES
>
> The whole test depends on that figure. If you enter the species in the tree instead of those in the group, the test undercorrects and you will "find" the false slowdown again. If you exaggerate the total, it overcorrects and you will lose a real slowdown.

8.4 The five models

The Fit the five models button compares, on the same branching times:

Table 8.1. The five diversification models.

Model	What it assumes
Pure birth (Yule)	A single speciation rate and **no extinction**.
Birth–death	Constant speciation and extinction, with the likelihood of Nee et al. for the reconstructed tree.
Density-dependent, logistic	The rate falls as niches fill up, toward a carrying capacity (Rabosky & Lovette 2008).
Density-dependent, exponential	The same, with an exponential decay instead of a logistic one.
Two rates	The rate changes once, at a moment that **is estimated** from the data.

They are compared by AIC with Akaike weights, as in Block 3: what you read is the difference from the best and the weight of the winner, not the absolute value.

> ! THE TWO-RATE MODEL ESTIMATES WHERE THE SHIFT IS
>
> And that is why it has an unfair advantage: it will always find the best possible place to put the jump, even in data generated with a constant rate. AIC penalizes the extra parameter, but not the fact of having searched for it. A high weight for this model is a clue, not a proof.

8.5 λ, μ, r and ε: what can be believed

The birth–death model estimates four related quantities:

Table 8.2. The four parameters, and how far they can be believed.

Symbol	What it is	Reliability
λ	Speciation rate	Reasonable.
μ	Extinction rate	**Very unreliable** with trees of living species only.
r = λ − μ	Net diversification	The best estimated, and the one you should report.
ε = μ/λ	Turnover	Poorly estimated, and it often comes out as 0.

> **AN E ESTIMATED AT ZERO DOES NOT MEAN THERE WAS NO EXTINCTION**
>
> It means that **the survivors' tree cannot see it**. It is the same problem as the e of biogeography in Chapter 10: the lineages that disappeared left no trace in the tips. With fossils, the estimate changes completely. Reporting r with confidence and ε with reservations is the honest thing to do.

The panel also gives the **doubling time**, ln(2)/r: how long the group takes to double its number of lineages at the estimated rate. It is the same information as r, stated in units you can picture—and compared with the age of the group, it tells you at once whether the rate explains the current diversity.

8.6 When there is only an age and a richness

Sometimes there is no tree of the group: only its age and how many species it has are known. The estimators of **Magallón and Sanderson (2001)** make do with that, and they are the ones used to compare hundreds of clades at once.

The panel asks for the number of `species`, the age, and—crucially—whether `the age is of` the **crown** or the **stem**.

> **CONFUSING CROWN WITH STEM**
>
> The **crown** is the most recent common ancestor of the living species of the group; the **stem** is its divergence from the sister group, and it is always older. With the stem, the same richness is spread over more time and **the rate comes out lower**. The two estimators are different and the program explicitly asks which one it is. A "group age" without a qualifier, taken from another paper, cannot be reused.

The estimator also needs an assumed value of ε, precisely because it cannot be estimated: the custom is to report the result with ε = 0 and with ε = 0.9, to show how much it depends on the assumption.

8.7 A rate for every species: DR

The previous models give one rate for the whole group. The **DR** statistic (Jetz et al. 2012) hands out a rate among the **tips**: for each species it weights the branches on the path to the root, giving more weight to the recent ones.

It is read **in relative terms**: which tips have high DR compared with the others **in the same tree**. The absolute values depend on the scale of the tree and are not compared across studies.

> **DR IS NOT A PER-SPECIES DIVERSIFICATION RATE**
>
> Even though it is very often used as one. It has been shown to estimate **speciation** better than net diversification, and to behave poorly when extinction is high (Title & Rabosky 2019). As a descriptive measure for ranking tips it is useful; as an estimator of r, it is not.

8.8 How well it does, measured

Checked against *ape* and *geiger*, with the results in the validation folder:

- The **γ** statistic gives **0.44734655**, identical to that of `ape::gammaStat`.
- The **pure-birth** model reproduces λ, its standard error and its log-likelihood identical to `ape::yule`.
- The **birth–death** model gives 0.24684678 against 0.24684645 from `ape::birthdeath`, and r = 0.20638381 against 0.20638489, with the same log-likelihood.
- The **Magallón and Sanderson** estimators agree with `geiger::bd.ms` to 10^{-7}, and the branching times to $1.2{\cdot}10^{-9}$.

> **A BUG THAT ONLY SHOWED IN THE BIAS**
>
> The tree simulator, used by the MCCR test, left the extinct lineages in—so more tips came out than requested, and the trees were not ultrametric—and, worse, **stopped the clock right at the last speciation** instead of drawing a present at random. That last point biased γ: it gave a mean of **+0.32** on constant-rate trees, where it should give 0, and would have bent every Monte Carlo test in the same direction. With the present drawn properly, γ gives a mean of −0.066 and a standard deviation of 0.9999, against 0.073 and 0.952 from `ape::rphylo`.

8.9 Common mistakes

> **REPORTING A NEGATIVE Γ WITHOUT THE MCCR TEST**
>
> It is the most common mistake of the block and the easiest to avoid. If you did not sample 100% of the group, your negative γ may be entirely an artifact of sampling. The MCCR test is two clicks away.

✕ INTERPRETING M OR E AS BIOLOGICAL QUANTITIES

From a tree of living species, the extinction rate is not estimated: it is assumed. Report r, say which ε you assumed, and show how the result changes with another.

✕ COMPARING DR ACROSS STUDIES

It depends on the scale of the tree and on the sampling. It serves to rank the tips of **your** tree, not to say that your genus diversifies faster than your neighbor's.

✕ ANALYZING A CHRONOGRAM WHOSE DATES DO NOT HOLD UP

This whole block inherits the uncertainty of Block 7. If the ages have intervals as wide as the tree, γ and the five models are describing noise. Before interpreting a slowdown, look at the intervals of the ages that produce it.

✕ TELLING A SLOWDOWN AS AN "ADAPTIVE RADIATION"

A curve that flattens is compatible with niche saturation, but also with incomplete sampling, with recent extinction and with a badly calibrated chronogram. The pattern does not identify the process, and the block measures the pattern.

9

CHAPTER NINE · BLOCK 9

Trait evolution

Two sister species are not two independent observations: they share almost all of their history. This whole block follows from that sentence. It fits three models of continuous evolution, measures phylogenetic signal in two ways that do not ask the same question, compares an ordinary regression with the one that corrects for relatedness, and reconstructs discrete ancestral states with their pie charts and with complete histories drawn on the tree.

9.1Why species are not independent data

9.2The character table

9.3Three models for a continuous character

9.4How much phylogenetic signal there is

9.5Two characters, one relationship

9.6A discrete character: the Mk model

9.7Ancestral states and stochastic mapping

9.8How well it does, measured

9.9Common mistakes

9.1 Why species are not independent data

Suppose you measure seed size in forty species and relate it to plant height. An ordinary regression treats those forty species as forty independent observations. They are not: if thirty belong to the same genus and inherited both the large seed and the height from their common ancestor, what you have is not thirty data points but, at bottom, **one** repeated thirty times.

> **THE PRACTICAL CONSEQUENCE**
>
> Ignoring relatedness **inflates significance**: p-values come out much smaller than they should, and relationships get published that do not hold up. The methods in this block are not a statistical refinement: they are the correction of a counting error.

You work on a tree, preferably the **dated** one from Block 7, because the models speak of rates per unit of time.

9.2 The character table

One row per species and one column per character, separated by commas, semicolons or tabs. The program infers the type of each column—continuous or discrete—, recognizes missing data and matches the names with the tree, **forgiving the difference between** `Bursera_palmeri` **and** `Bursera palmeri`, which is the one that wastes the most time.

Whatever does not match is **listed**, not silently swallowed. And each analysis **prunes the tree** to the tips that have the character it needs, and says how many it kept.

9.3 Three models for a continuous character

Table 9.1. The three models, in the parameterization of *geiger*.

Model	What process it describes
Brownian motion (BM)	Pure drift: the character wanders at random, and the variance between two species grows with the time since they split. A single parameter, σ^2.
Ornstein–Uhlenbeck (OU)	A force, α, that pulls the character toward an optimum. It is the model of stabilizing selection.
Early burst (EB)	The rate **decays** with time: a lot of change at the beginning and little afterward. It is the model of adaptive radiation.

They are compared by AIC with Akaike weights, and the panel writes the verdict in words.

The half-life, instead of α

OU's α is hard to interpret by eye. The program reports the **half-life**, ln(2)/α: the time the character takes to travel half the way toward its optimum. And it is read **by comparing it with the depth of the tree**:

- Half-life much shorter than the depth: strong stabilizing selection, and the tree hardly matters for predicting a species' value.
- Half-life longer than the depth: OU is **indistinguishable** from Brownian motion, and its α means nothing.

> **! TWO WARNINGS ABOUT OU**
>
> First: OU and early burst **assume contemporaneous tips**. On a non-ultrametric phylogram their parameters do not mean what they say, and the block measures the spread of the tip heights and warns you.
>
> Second: with fewer than fifty species OU is often preferred for **statistical, not biological** reasons. It is a model very good at fitting noise, and winning by AIC is not the same as describing a real process (Cooper et al. 2016).

When α hits the limit of the range explored, the card prints "—" instead of a number and says that α ended up at the limit: a number there would be an invention.

The traitgram

The figure that puts the character and the tree **in the same plane**: time on one axis, the value of the character on the other, and the branches of the tree drawn between the reconstructed values. It shows at a glance whether there was convergence, and the tip labels are spread out so they do not overlap.

9.4 How much phylogenetic signal there is

"Phylogenetic signal" is how much more relatives resemble each other than would be expected by chance. The two usual measures **do not ask the same question**, and that is why they can seem contradictory without being so.

Table 9.2. The two measures of phylogenetic signal.

Measure	What it manipulates	How it is read
Pagel's λ	The **shape** of the tree: it multiplies the covariances between species without touching the variances.	λ = 1 is pure Brownian motion; λ = 0, a character distributed as if the phylogeny did not exist. It is tested against 0 and against 1 with a likelihood ratio.
Blomberg's K	The **magnitude** of the variance: it compares the observed one with the one Brownian motion predicts on that tree.	K = 1 is exactly Brownian; K > 1, relatives resemble each other **more** than Brownian motion expects. It comes with a permutation test.

K < 1 WITH λ ≈ 1 IS NOT A CONTRADICTION

They are different questions about the same data: λ looks at the **shape** of the tree and K at the **magnitude** of the variance. A character can be distributed exactly as the tree predicts—λ = 1—and at the same time vary less among species than Brownian motion expected—K < 1. The panel explains this when it happens, because it is one of the most puzzling readings.

The plot of **K under the null hypothesis** draws the distribution of K when the values are shuffled among species, and places the observed one on it. Without that plot, a K of 0.8 does not tell you whether it is a lot or a little.

SIGNAL IS PATTERN, NOT PROCESS

λ = 1 **does not prove** neutral drift, and a low λ may be due to measurement error and not to biology. With fewer than twenty species, moreover, both measures are estimated with very little precision.

9.5 Two characters, one relationship

The panel fits **three** regressions on the same two characters and puts them side by side:

1. The **ordinary** one, which counts each species as a new data point.
2. Felsenstein's **independent contrasts**: they turn the values of *n* species into *n*−1 differences between sister nodes, each scaled by its branch lengths. Those differences **are** independent, and their regression is fitted **through the origin**, because the sign of a contrast is arbitrary and an intercept would mean nothing.
3. **PGLS**, which solves the same problem by another route: a generalized regression whose covariance matrix is that of the tree.

THE GARLAND AND IVES IDENTITY

Contrasts and PGLS **are not two similar approximations**: under Brownian motion they give **exactly the same slope**. The program checks this in the interface itself and shows it: the slope of the contrasts through the origin and that of PGLS with λ = 1 have to be the same number. When they are not, something is wrong, and it is a check worth looking at.

Comparing the p-value of the ordinary regression with that of PGLS shows, at a single glance, how much the significance was being exaggerated.

> ✕ BELIEVING THAT PGLS FIXES OTHER THINGS
>
> It corrects **non-independence**. It does not correct causality, an omitted variable or measurement error. And if the phylogenetic signal of the residual turns out to be nil, PGLS and least squares give the same thing: the correction does not always change the conclusion, and that too is a result.

9.6 A discrete character: the Mk model

Lewis's (2001) Mk model treats the states of a character the way a substitution model treats nucleotides: a matrix of transition rates between states. Three variants:

- **ER**: a single rate for all changes.
- **SYM**: the rate from i to j equal to that from j to i.
- **ARD**: all different. With k states there are $k(k-1)$ parameters: with four states, twelve.

They are compared by AIC. With few species ER almost always wins, and not because the biology is simple: because there are not enough data to estimate more.

> ✕ NOT CHECKING WHETHER THE RATE BLEW UP
>
> A rate that goes to infinity is the symptom of a **saturated** character: it has changed so many times that the tree no longer keeps any memory of its history. The nodes all come out at 0.5 and the reconstruction says nothing. That is not a failure of the program: it is the data saying they do not know.

> ℹ A CONVENTION WORTH KNOWING
>
> The log-likelihood reported by `ape::ace` is that of `phytools::fitMk` **plus log(k)**, and the whole difference is the flat prior at the root. It is not that one of them is wrong: they are two conventions. The program reports both, so you can compare with either package without surprises.

9.7 Ancestral states and stochastic mapping

The pie charts

Each internal node carries a pie chart with the probability of each state. The version reported is the **marginal** one, which integrates over everything else in the tree. A pie that is almost a single color is a firm reconstruction; one that is split means that **the data do not know**, and that split **is the result**, not a defect of the drawing.

The program draws the pies on purpose, instead of writing the winning state: that way a node at 55% **looks** like what it is. Uncertainty grows toward the root, which is often exactly the node of interest.

Stochastic mapping

Instead of giving a probability per node, it draws **complete histories** of the character on the tree—where and when each change happened—consistent with the tips and with the model (Huelsenbeck et al. 2003). Five hundred histories in less than a second.

What is reported is the **average over the histories**: the expected number of changes, how much time the tree spent in each state, and how often each branch contains a change. That is considerably more informative than a single state per node: it turns "the ancestor was probably yellow" into "the lineage spent 70% of its history in yellow and there were about four changes."

> **THE HISTORIES ARE ONLY AS GOOD AS THE MK THAT GENERATES THEM**
>
> If the rates are badly estimated, **all** the histories will be off in the same direction, and the average does not reveal it. The spread among histories measures the uncertainty *within* the model, not that of the model.

9.8 How well it does, measured

This block is the one with the most reference implementations to face, and it was checked against all four: **geiger**, **ape**, **nlme** and **phytools**, with **42 checks** and no disagreement.

- The **independent contrasts** reproduce `ape::pic`.
- **PGLS** is identical to `nlme::gls` in slope, standard error, t and p.
- Its slope matches that of the contrasts through the origin, just as the **Garland and Ives identity** requires.
- The **marginal states** match `ape::ace` and `phytools::ancr`, which in turn agree with each other.
- The three continuous models reproduce the parameterization of `geiger::fitContinuous`.

> **THREE BUGS THE VALIDATION UNCOVERED**
>
> **Early burst lost all its digits** when the parameter was small: the expression $(e^{a \cdot s}-1)/a$ was computed as written, and for a tiny |a| that means subtracting two nearly equal numbers. It was replaced by the function that exists precisely for that.
>
> **The t distribution function came out symmetric about zero**, because the incomplete beta only sees t^2. The p-values were correct by sheer luck, because it was always called with |t|.
>
> **The OU α was bounded between fixed numbers** instead of numbers relative to the depth of the tree, and it once reported a half-life of **691,179 substitutions** on a tree 0.08 deep. Now the limit is relative, and if α hits it, the card says so instead of printing the figure.

9.9 Common mistakes

✕ PUBLISHING AN ORDINARY REGRESSION ACROSS SPECIES

It still appears in journals. If the data are species and a phylogeny is available, the ordinary regression miscounts and its p-value is too small. The panel shows the three regressions together precisely so the difference can be seen.

✕ FITTING OU ON A NON-ULTRAMETRIC TREE

OU and early burst assume that all tips are contemporaneous. On a phylogram, their parameters do not mean what they say. The block warns you; you have to read the warning.

✕ CONCLUDING "STABILIZING SELECTION" BECAUSE OU WON

With few species OU often wins for statistical reasons. Look at the **half-life** against the depth of the tree: if it is longer, OU is indistinguishable from Brownian motion and there is nothing to conclude.

✕ REPORTING THE WINNING ANCESTRAL STATE WITHOUT ITS PROBABILITY

"The ancestor was yellow" is very different from "the ancestor was yellow with probability 0.55." The pie charts are drawn so that this difference cannot be hidden, and the complete figure is what gets published.

✕ FORGETTING THAT THE TREE WAS PRUNED

Each analysis uses only the tips that have the character it needs. If out of eighty species only forty have seed size measured, your λ was estimated on forty. The panel says so; you should report it too.

10

CHAPTER TEN · BLOCK 10

Historical biogeography

A geographic range is a character too, but with one difference that changes everything: when a lineage splits, the range **is divided** between the two daughters, and how it is divided is exactly what separates one model from another. This block fits six of them, reconstructs where each lineage was—including the corners that almost nobody draws—and turns the probabilities into counts of dispersals, extinctions and splits.

10.1The character that splits in two

10.2The areas and who lives in which

10.3How many ranges a lineage can have

10.4The six models

10.5Why +J almost always wins

10.6Where each lineage was, and the corners

10.7How many times each thing happened

10.8How well it does, measured

10.9Common mistakes

10.1 The character that splits in two

In Block 9 a discrete character changes along a branch and, on reaching a node, both daughters inherit whatever the parent had. A geographic range does not behave like that. If the ancestor lived on islands A, B and C and the lineage splits, the usual outcome is **not** that both daughters inherit all three islands: the usual outcome is that one keeps A and the other BC, or that one keeps everything and the other a single island.

That forces two things at once. First, the **states** of the model are not the areas but the **subsets** of areas. Second, two different processes have to be described: what happens to the range **along a branch** and what happens to it **at the node**.

Table 10.1. The two processes and their parameters.

When	What happens
Along a branch (anagenesis)	At rate **d**, the range **gains** an area: dispersal. At rate **e**, it **loses** one: local extinction. A single-area range that loses its area **disappears**, which is why the null range is a state, and an absorbing one.
At the node (cladogenesis)	The ancestral range is divided between the two daughters. **Which divisions are allowed** is the only thing that distinguishes the three base models.

> **FIRST YOU NEED A TREE, AND IT HAD BETTER BE DATED**
>
> The dispersal and extinction rates are estimated **per unit of time**. On a phylogram measured in substitutions they do not mean what they say. The panel sends you to Block 7 if you do not have a chronogram yet, and rightly so: a d of 0.035 only means something if the axis of the tree is in millions of years.

10.2 The areas and who lives in which

The program reads the **LAGRANGE format**, which is the same one BioGeoBEARS uses: a header line with the number of taxa, the number of areas and their names in parentheses, and then one row per species with zeros and ones.

It also accepts a **comma-separated table** with one column per area, which is easier to put together in a spreadsheet, and it exports in LAGRANGE format in case you want to take the same data to another program. The bundled example is of *Bursera*, and it lets you walk through the whole block without bringing anything of your own.

The names have to match the tips of the tree. The block says how many species it kept and which ones it did not find, instead of carrying on in silence.

10.3 How many ranges a lineage can have

With n areas there are 2^n subsets. With five areas that is 32 states; with eight, 256. The transition matrix grows with the square of that, and that is where the analysis becomes impossible.

The remedy is the **maximum range size**. If no living species occupies more than three areas, limiting the states to ranges of three areas or fewer cuts the problem down enormously.

> **! THE CAP IS NOT A TECHNICAL SETTING: IT IS AN ASSUMPTION**
>
> A lineage that was in five of six areas **cannot be reconstructed** if you set the cap at three. You have decided in advance that it did not happen. Often that is reasonable—very wide ranges are rare—but it is a biological decision, and it has to be reported along with the results.

The null range

It comes switched on. It is the state a lineage reaches when it loses its only area, and it is **absorbing**: there is no way out of it. Having it is what makes extinction a real process inside the model and not just a label.

The dispersal multipliers

The matrix of multipliers between areas is, of everything in the block, what most resembles doing real biogeography. Each cell multiplies the dispersal rate *from* one area *to* another: a 0 forbids it, a 0.1 makes it difficult, a 1 leaves it like the rest. **That is where an isthmus that closed, a land bridge that opened or an ocean that was never crossed comes in.** You edit it by clicking a cell, and there is one button to set both directions at once and another to put everything back to one.

> **i GEOGRAPHY THAT CHANGES THROUGH TIME**
>
> The engine knows how to work with **time strata**—a branch that crosses a boundary gets the product of the matrices of the periods it passes through, each with its own multipliers and its own existing areas—but the interface of this block does **not yet expose** those controls. What you can edit today is a single matrix, constant over the whole tree.

10.4 The six models

Three base models, each with and without **+J**. All three share d and e; what changes is which divisions they allow at the node.

Table 10.2. The three base models and which division they allow when a lineage splits.

Model	Divisions allowed
DEC Ree and Smith 2008	Narrow sympatry (a single-area ancestor is copied into both daughters), subset sympatry (one daughter keeps the whole range and the other a single area) and **narrow vicariance**: the range is cut in two, but one of the daughters keeps **a single** area.
DIVALIKE	Only vicariance, but **with any cut**: the range is split into two complementary halves of whatever size. There is no subset sympatry.

Table 10.2. The three base models and which division they allow when a lineage splits. *(continued)*

Model	Divisions allowed
BAYAREALIKE	Neither of the two: **both daughters inherit the whole range**. The geographic division does not happen at the node; if it happens, it is afterwards, by dispersal and extinction along the branches.
+J Matzke 2013	Added to any of the three: the **founder event**, a daughter that leaves for a single area **outside** the ancestor's range. It is the model of long-distance colonization.

You can fit all six, only the three without J, or only DEC and DEC+J. They are compared by **AICc with Akaike weights**, and each +J model also gets a likelihood-ratio test against the model it contains.

> **I** IF E COMES OUT AS ZERO
>
> The panel spells it out: an extinction rate estimated at zero **does not mean there was no extinction**; it means that no trace of it was left in this tree. With small trees that is the norm.

10.5 Why +J almost always wins

It is the most delicate point of the block, and the program does not hide it: **whenever a +J model wins, the panel explains why that is to be expected and proves nothing on its own.**

> **∑** REE AND SANMARTÍN'S OBJECTION (2018)
>
> The comparison is not clean. The parameter **j buys probability at cladogenesis**, which is an instantaneous event and costs nothing along the branches; **d and e pay for theirs** by integrating over the whole time of the tree. They are not competing on equal terms.
>
> The fact that clinches the argument: in the cases they examined, **+J won even on data simulated without any jump**.

What can be stated, and what the block puts in the table, is **by how much** it wins and **over which model**. The likelihood ratio is reported with the warning that **j lies on the boundary of the parameter space**, and a p computed that way is conservative.

10.6 Where each lineage was, and the corners

Once a model is chosen, the **marginal** ancestral ranges are reconstructed. Each node carries a pie with the ranges that have some probability; everything below the threshold—0.05 by default, and it can be changed—is lumped into a gray slice, so that the drawing does not become unreadable with thirty ranges of probability 0.001.

The corners

This is the part that many published figures leave out. A pie on the node says what range the **ancestor** had; it does not say **what each daughter got**, which is precisely the biogeographic event. The small boxes on the two daughter branches—the **corners**—show the range each one starts with immediately after the split.

> **THE HALF OF THE STORY THAT USUALLY GETS LOST**
>
> Knowing that the ancestor was in ABC does not tell you whether there was vicariance AB|C, subset sympatry ABC|A or a jump to D. That distinction **is in the corners**, and it is what gets told in the paper. They come switched on; leave them on.

10.7 How many times each thing happened

One probability per node does not say how many dispersals there were or how many times a range was split. **Biogeographic stochastic mapping** draws complete histories—a range for every node, a division for every split and the moves along every branch—conditioned on what is seen at the tips, and by repeating it turns the reconstruction into **counts of events**, which is what a biogeographic history is made of. Five hundred replicates by default.

What is reported, averaged over the histories, is six quantities:

- **Dispersals** and **local extinctions** along the branches.
- The internal nodes divided among the **four types of split**: sympatry, subset sympatry, vicariance and founder event.

And with them, two warnings worth reading. If the **standard deviation** of the dispersals is large relative to their mean, the data do not pin down how many there were: only the order of magnitude. And if the sampling ran out of attempts on some branch, the panel says **how many times** that happened and warns that it slightly underestimates the counts—it tends to happen with long branches and many areas—instead of returning a clean number that would be wrong.

Everything comes out in four files: the model table, the ancestral ranges, the mapping events and the areas in LAGRANGE format.

10.8 How well it does, measured

It was validated against **BioGeoBEARS 1.1.3** on the Hawaiian *Psychotria*: the data set that LAGRANGE and BioGeoBEARS distribute, and the one that every paper discussing DEC+J analyzes again. **41 checks** and not one disagreement.

Table 10.3. Log-likelihoods of the six models on the Hawaiian *Psychotria*.

Model	PhylogenyPro	BioGeoBEARS 1.1.3
DEC	−34.54195625	−34.54195758
DEC+J	−20.94758871	−20.94758855
DIVALIKE	−33.14967582	−33.14967538
DIVALIKE+J	−21.08620942	−21.08620912
BAYAREALIKE	−40.33442556	−40.33442795
BAYAREALIKE+J	−21.55262993	−21.55262941

The rates d and e agree to about $1{\cdot}10^{-5}$ (in DEC, 0.03505 and 0.02831 against 0.03505 and 0.02836), and the **ancestral ranges** agree to $\mathbf{1.75{\cdot}10^{-8}}$ at the eighteen internal nodes when both are evaluated with the same parameters; the root comes out as KO with 0.51655966 against 0.51655968.

THE ROOT CONVENTION

LAGRANGE and BioGeoBEARS weight every allowed range at the root by one, and that is what **every published DEC log-likelihood** means. A truly flat prior differs by exactly the logarithm of the number of ranges—here, **2.708050**. It is the same constant for every model on the same areas, so no comparison changes; but the number a reader checks against a paper does. The program reports the BioGeoBEARS convention and says so.

WHAT TURNED UP DURING VALIDATION

The reference was wrong before the code was. DIVALIKE disagreed by 0.033, and the cause was in the R script: BioGeoBEARS reads the parameter `mx01v`, not `maxent01v`. Setting the latter does absolutely nothing and leaves DIVALIKE without its 2–2 vicariances, which amounts to not being DIVALIKE.

The upward message in the reconstruction included the daughter's own likelihood, counting it twice. The symptom is the one to remember: reconstructions that are **too confident** and biased toward wide ranges—a node at 0.96 for KOMH where the right answer was 0.11. The message passed down to a daughter has to carry everything **except** that daughter's subtree; the corner probability, which is the one you read, does use all the data. They are two different quantities, and now both are computed.

And speed, three times over. Fitting one model took 7.4 s. Reusing the buffers inside the matrix exponential brought it down to 0.3 s—the garbage collector had been doing almost all the work; tying the optimizer's starting values to the depth of the tree removed nearly all the remaining iterations; and replacing the per-branch exponential with **uniformization** left it at **0.1 s, some seventy times faster**, without a single digit changing. Since every term of that sum is non-negative, it also can no longer produce the slightly negative probabilities that the Taylor series did.

10.9 Common mistakes

CONCLUDING "THERE WAS LONG-DISTANCE DISPERSAL" BECAUSE +J WON

It is the central mistake of this block, and that is why the panel warns about it on its own. +J almost always wins, even with data simulated without a single jump. Report by how much it wins and over which model; do not turn it into a biogeographic conclusion.

SETTING A LOW RANGE CAP AND NOT SAYING SO

The cap decides in advance that no lineage was ever in more areas than that. It is a strong assumption and cheap to report: one sentence in the methods is enough.

RUNNING IT ON A PHYLOGRAM

d and e are rates per unit of time. On a tree in substitutions per site the numbers come out, but they do not mean dispersals per million years. Use the chronogram from Block 7.

PUBLISHING ONLY THE NODE PIES

The node says where the ancestor was; the **corner** says what each daughter got, which is the event. Without the corners, the figure cannot tell a vicariance from a subset sympatry.

READING A SPREAD-OUT PIE AS A FAILURE

As in Block 9: a node with five ranges at 0.2 **is** the result. The gray slice is not noise in the way; it is the part of the range space that the data do not rule out.

LEAVING EVERY MULTIPLIER AT ONE AND CALLING IT A MODEL

With a flat matrix you are claiming that moving between two neighboring islands costs the same as crossing an ocean. Sometimes that is the null hypothesis you want; it is almost never the story you want to tell.

11

CHAPTER ELEVEN · BLOCK 11

Comparing trees

Every earlier block produces a tree. This one asks what to do when two of them are not the same, which with real data is always. It measures distances that do not break in the same way, draws the disagreement instead of hiding it, separates **being well estimated** from **being well supported**, builds a species tree from the quartets and tells apart—as far as that is possible—incomplete lineage sorting from introgression.

11.1The trees to be compared

11.2How alike they are

11.3The tanglegram, and its trap

11.4Concordance is not support

11.5A species tree from the quartets

11.6Drawing the conflict instead of resolving it

11.7The ABBA-BABA test

11.8How well it does, measured

11.9Common mistakes

11.1 The trees to be compared

Any combination will do: the ones made by Blocks 4, 5, 6 and 7, and the ones you paste in Newick, one per line —the trees of each gene separately, a colleague's, a paper's. At least two are needed.

They all have to have the same tip names, and whatever does not match **is named**. It is one of those checks that look like red tape until the day an extra underscore has you comparing fifteen species while believing you are comparing sixteen.

11.2 How alike they are

The menu offers six measures, all on the **unrooted** tree, and the block computes them for every pair and puts them in a table and in a heat map.

Table 11.1. The distance measures between trees.

Measure	What it counts
Robinson–Foulds	How many splits one has and the other does not. Topology only.
Normalized Robinson–Foulds	The same, divided by its maximum possible value, so that trees of different sizes can be compared.
Disagreeing quartets	Of all the groups of four tips, how many are resolved differently in one tree and in the other.
Weighted Robinson–Foulds	Like the first, but each split weighs as much as its branch is long.
Branch score Kuhner and Felsenstein	The difference between the branch lengths of the shared splits, summed in squares.
Path difference Steel and Penny	How much the distance between each pair of tips changes when going from one tree to the other.

> ✦ THEY DO NOT BREAK IN THE SAME WAY, AND THE BLOCK SAYS SO ALONGSIDE
>
> **A single misplaced tip can take Robinson–Foulds to its maximum**, while the quartet distance barely moves. When that happens, the disagreement **is one tip, not one history**, and confusing the two is the costliest mistake in this section.
>
> The practical reading: if RF comes out huge and the quartet distance small, look for the rogue taxon before writing that the two analyses give different histories.

11.3 The tanglegram, and its trap

Two trees facing each other with one line per tip. The lines cross exactly where the trees disagree, so the figure shows **where** the disagreement is and not just how much of it there is.

Before reading anything you have to **untangle**, and it comes switched on. Untangling means **rotating branches** around their nodes until the lines cross as little as possible. A rotation **does not change either tree**: it is the same topology drawn another way. Without that step, a tanglegram of two identical trees can look like a ball of yarn.

> **A TANGLEGRAM WITH NO CROSSINGS DOES NOT MEAN THE TREES ARE EQUAL**
>
> It is the warning that almost no program prints, and the block carries it. A rotation can line up **two different topologies**: the lines come out parallel and the eye concludes "they are the same tree." That is what the distances of the previous section are for. The tanglegram says **where**; the number says **how much**.

11.4 Concordance is not support

Bootstrap support says how confidently **those data** hold up a branch. It does not say how many genes see it. They are two different questions, and with whole genomes the second one often matters more.

The **concordance factors** of Minh and colleagues (2020) answer the second one, branch by branch:

- **gCF**: what percentage of the **gene trees** contains that branch.
- **sCF**: what percentage of the alignment's **sites** supports it, estimated on quartets drawn at random around the branch—a hundred per branch by default.

They are drawn as stacked bars against the one third that **pure chance** would give, which is the right reference when there are three possible resolutions, and as a scatter of gCF against sCF.

> **WELL ESTIMATED AND POORLY SUPPORTED AT THE SAME TIME**
>
> A branch with **100% bootstrap and 35% concordance** is not a contradiction: it means that the data you have hold it up without hesitation **and** that two out of three genes see something else. It is well estimated and poorly supported at the same time, and both halves of that sentence have to appear in the paper. The block repeats it where needed.

> **THE DEEPEST BRANCH**
>
> A branch that hangs directly from a bifurcating root **has nothing "above"**: the root is not a node of the unrooted tree. The four groups around the branch are then formed from the daughters of the sister. It is an easy case to overlook—and we overlooked it, as §11.8 tells—but today's count handles it, and that is why the concordance table includes **all** the internal branches and not all but the deepest. It is worth looking at: the deepest branch is often the one that separates the two large groups of the study, that is, precisely the one discussed in the abstract of the paper.

11.5 A species tree from the quartets

When genes disagree because of **incomplete lineage sorting**, concatenating them fixes nothing: it can give, with great confidence, a tree that no gene supports. The way out is to work with the quartets.

1. For every group of four species there are **three** possible resolutions. The program counts how often each one appears among the gene trees.
2. Each candidate tree gets a **score**: how many of the observed quartets it can explain.
3. The best one is sought with a nearest-neighbor search, starting from the tree you choose.

> **IT IS A HEURISTIC, AND IT CALLS ITSELF A HEURISTIC**
>
> The search can get stuck in a local optimum. **It is not the exact ASTRAL program**, and the block does not pretend it is. For a paper on gene discordance, use ASTRAL; this is for seeing the problem, understanding it and comparing with what you already have.

11.6 Drawing the conflict instead of resolving it

A consensus has to **throw away** the splits that contradict each other. A network **keeps them all** and draws them as boxes: the wider the box, the more weight the conflict carries. Two routes:

Table 11.2. The two networks the block builds.

Construction	What it starts from and what it shows
Consensus network of the trees	Takes the splits that appear in enough trees—the threshold is set at 0.3—and draws them all. The **incompatible** pairs are found and marked in color: those are the boxes.
Circular decomposition of the distances	Starts from a distance matrix instead of trees. If you give it the distances of a tree, **it returns that tree exactly**; if you give it anything else, the difference shows up as width.

> **WHY DRAWING THE CONFLICT IS WORTH IT**
>
> A consensus with polytomies says "I don't know here." A network says "here there are **two** signals, and these are they." It is much more information, and it is honest: it does not force a choice too early.

11.7 The ABBA-BABA test

With the tree (((P1,P2),P3),O), a site where **P2 and P3** share the derived state—**ABBA**—and another where **P1 and P3** share it—**BABA**—are equally incongruent with that tree. And under incomplete lineage sorting they are **equally likely**.

That is the whole idea: incomplete lineage sorting is symmetric, and introgression is not. One of the two patterns being much more frequent than the other is the signature of hybridization. **Patterson's D** measures that imbalance, and the block gives it a 95% interval by **block jackknife**—twenty blocks by default—because neighboring sites are not independent and a standard deviation computed site by site comes out too small.

You can test one particular trio or sweep **all** the trios against an outgroup.

> **D ≠ 0 HAS THREE EXPLANATIONS AND ONLY ONE IS INTROGRESSION**
>
> The same imbalance is produced by a **badly chosen outgroup**, **very unequal rates of evolution** between P1 and P2, and an **alignment error**. All three have to be ruled out before writing "hybridization." The panel says so every time, not once in the help.

Everything goes out in five files: the distance matrix, the concordance per branch, the splits of the network, the ABBA-BABA results and all the trees together in `.tre`.

11.8 How well it does, measured

It was checked against **ape 5.8.1** and **phangorn 2.12.1**: **25 checks** and not one disagreement.

- **The distances on 36 pairs each**, 216 numbers: RF and normalized RF **exact**; weighted RF to $4.5 \cdot 10^{-10}$, branch score to $1.5 \cdot 10^{-10}$, path difference to $5 \cdot 10^{-14}$ and its weighted form to $8 \cdot 10^{-10}$.
- **The quartet distance** against a brute-force count over the 1,820 quartets of sixteen taxa, pruning both trees to each one: 1,767/53, 1,691/129 and 1,820/0, **exact**.
- **gCF** against the same definition recomputed in R over the thirteen internal branches, with a difference of **zero**.
- **The consensus network** split by split against `phangorn::consensusNet`: the thirteen non-trivial splits, identical.
- **Patterson's D** against the definition computed in R on an alignment of 60,000 sites with 15% introgression: ABBA 1,089, BABA 753, D 0.18241042, jackknife standard error 0.07884390 and z 2.31356419, **every digit**.

> **WHEN THERE IS NO PACKAGE TO COMPARE AGAINST**
>
> For the **circular decomposition** none is installed, so it was checked against an **identity**: fed with the path distances of a sixteen-taxon tree it returns exactly its 13 internal splits and its 16 terminal ones, with the branch lengths to $3 \cdot 10^{-16}$ and the reconstructed distances to $9 \cdot 10^{-16}$, without trimming anything; and the outer cycle of the network closes on itself to $3 \cdot 10^{-16}$. An identity that has to hold is as good a test as a package, and sometimes better.

WHAT TURNED UP DURING VALIDATION

The weighted measures read half a branch. A rooted tree has two edges below its root, and they are **a single** edge of the unrooted tree; taking the length of whichever node the split map had kept cut it in half. Every weighted comparison inherited the error: weighted RF came out as 2.54 where phangorn says 3.78. Now the lengths are accumulated **per split**, not read from a node.

The path difference counted the root. Measured in edges, the path between two tips depends on where the root is, and two identical trees rooted differently came out at 7.94 instead of 0. The measure belongs to the unrooted tree, so now the first thing done is to remove the root.

A tree rooted on a single tip grew one internal branch too many. The branch above its sister contains n−1 tips, and the test for "trivial" only looked at the lower side. One of the app's own example trees showed 16 internal splits where there are 15, which inflates by one every comparison with a tree rooted elsewhere. **It turned up when running the block on real output**, not in the benchmark, whose reference trees are never rooted that way.

The four groups around a branch were three below the root, for the reason given in §11.4: exactly one branch, always the deepest, was missing from every concordance table. Twelve branches where there are thirteen.

The circular decomposition had its four terms paired the wrong way round. The weight of a split is a four-point difference across the **two gaps** that bound it, and pairing the distances by gap instead of by end gives a number that looks like a weight and is not one: 91 splits where a sixteen-taxon tree has 29, and distances it could not reproduce.

And a small one with a big moral: `opts.minSites || 20` turned a deliberate zero into twenty, so asking for the sweep of all the trios returned none.

11.9 Common mistakes

REPORTING A LARGE ROBINSON–FOULDS AS "DIFFERENT HISTORIES"

Look at the quartet distance first. If that one is small, what you have is **one unstable tip**, not two conflicting hypotheses. Reporting both measures costs one line and avoids the wrong conclusion.

READING A TANGLEGRAM WITH NO CROSSINGS AS PROOF OF EQUALITY

A rotation can line up two different topologies. The tanglegram shows where they disagree; what says whether they disagree is the number.

✕ CONFUSING CONCORDANCE WITH SUPPORT

100% bootstrap and 35% gCF describe different things, and both are true. Publishing only the first gives a very optimistic idea of what you know.

✕ CONCATENATING EVERYTHING WHEN THE GENES REALLY DISAGREE

Under incomplete lineage sorting, concatenation can give, with very high confidence, a tree that **no gene** supports. That is where the quartets are needed.

✕ ANNOUNCING HYBRIDIZATION WITH A SIGNIFICANT D AND NOTHING ELSE

Rule out the outgroup, the unequal rates and the alignment first. A significant D is the **beginning** of the investigation, not its conclusion.

✕ PRESENTING THE QUARTET TREE AS IF IT WERE ASTRAL

It is a heuristic search and it can get stuck in a local optimum. Say so in the methods, or run ASTRAL.

12

CHAPTER TWELVE · BLOCK 12

Figures and report

A result that cannot leave the program is not a result. Here the figure is built—with all its decisions in view and reversible—, the methods are written with the numbers of the run and not those of a template, and everything is packed so that somebody else can repeat it. It is the block that delivers the work.

12.1The tree studio

12.2Cutting the tree into groups

12.3Pictures of the OTUs and the clades

12.4Labels that do not overlap

12.5The three tree formats

12.6The figure studio

12.7The resolution the journal asks for

12.8The methods, written with the numbers of the run

12.9The reproducible package

12.10How well it does, measured

12.11Common mistakes

12.1 The tree studio

The figure of a tree involves more decisions than it seems, and **almost all of them change what the reader understands**: whether it is drawn with branch lengths or without them, which support is shown and from what threshold, where the root goes, in what order the branches are ladderized. All of them are changed here and the drawing is redone on the spot.

Table 12.1. The decisions that most change how the figure is read.

Control	Why it matters
Cladogram (no lengths)	A cladogram only asserts relationship. A phylogram also asserts **how much change** there was on each branch. Drawing one and talking about the other is the most common misunderstanding of a phylogenetic figure.
Support and threshold	Hiding low support values from the figure is legitimate if you say so; hiding them and not saying so turns a doubtful tree into a solid one to the eye.
Root and outgroup	The root does not come from the data: it is placed. Changing it changes which group is sister to which, and it is the decision with the most consequences in the whole drawing.
Ladderize	Orders the branches by size. It is purely aesthetic—it does not change the tree—and it makes legible a figure that otherwise looks like a mess.
Scale bar	Without it, a phylogram does not say what units it is in. With a chronogram, the axis already says so.

The rest is finish: width, row height, tip font and its weight, italics for scientific names, title, alignment of the names with their dotted leader, and color for the clades you mark by hand.

12.2 Cutting the tree into groups

You ask for a number of groups, *k*, and the program cuts the tree in one of three ways:

- At the **longest branches**, which works on any tree.
- By **depth**, which only makes sense on an ultrametric tree: if yours is not one, **it says so and switches method** instead of returning a meaningless cut.
- From **clades** or by hand.

The color can go on the branches, on the names, on a side strip or on soft background bands, with its legend and with the group names you set.

IT SAYS WHETHER EACH GROUP REALLY IS A CLADE

Next to the sizes, the panel reports the **monophyly** of each group. Coloring a paraphyletic group and saying nothing is one of the neatest ways for a figure to mislead: the reader sees a color and understands "lineage," when what is there is a cut of convenience.

And the background bands are drawn only for groups whose tips are **contiguous** in the figure, because a band spanning tips of another group would be claiming that they belong to it.

12.3 Pictures of the OTUs and the clades

You can hang photographs or drawings on each tip, each group or the clades you mark, at whatever size you want, as circles, squares or rounded squares, with or without a frame. The pictures are dragged into the manager, and **a file named after its taxon finds its place on its own**.

Each picture has its crop, its zoom, its rotation and its brightness, contrast, saturation, temperature and sharpness settings, plus a **background remover** that floods inward from the edge—so that light areas *inside* the object survive, which a brightness threshold would destroy. Each one carries its credit and its license.

NOTHING IS DRAWN UNTIL YOU CONFIRM THAT THE PHOTO IS OF THAT SPECIES

It is a deliberate checkbox and it cannot be skipped. **A photograph hung on the wrong branch is a claim, not an ornament**, and it is a claim the reader is going to believe without checking it.

WHERE THEY LIVE AND WHERE THEY TRAVEL

The pictures are stored in the browser—in IndexedDB, separated by data set—and **never leave your computer**. On export, they travel **inside** the SVG, so the figure does not depend on any loose file that gets left behind when the paper is sent.

12.4 Labels that do not overlap

The program does two things when labels get in each other's way. The first is to **separate** them: push them along an axis until none is closer to another than the requested spacing, without leaving the frame. The second, when no more will fit, is to **drop the least important ones** and keep the ones that say the most.

AND IT SAYS HOW MANY IT DROPPED

That is the part that matters. Losing labels silently is worse than not drawing them: the reader does not know they are missing. The panel says **how many it hid and where to find them**—they are always in the table and in the package—and what to change if you want them all.

In the unrooted layout nothing prevents two tips from landing side by side, so there the ones that still collide are dropped and the most peripheral are kept, again with the count in view.

12.5 The three tree formats

Table 12.2. The three formats and what each one is for.

Format	What it carries
Newick	Universal and narrow. Everything reads it, and that is why it is the one to send when only the tree is needed.
NEXUS	With its `TRANSLATE` block and the block that **FigTree** reads.
phyloXML	The only one of the three that can carry at once the **support** of a node, its **age** and its **confidence interval**, which is exactly what Blocks 5, 6 and 7 produce. If your tree is dated, it is the format that loses nothing.

You choose whether branch lengths go in, whether support values go in, with how many decimals, and whether only the chosen tree is written or all of them. It is downloaded or copied to the clipboard.

12.6 The figure studio

Whatever is changed here applies to **every figure in the twelve blocks**, not just the tree: palette, font family and size, stroke width and background. The ten colors can be edited one by one—they are the ones used by the groups of the tree, the series of the curves, the states of a character and the areas of the biogeography—and separately there are those of the text, the axes and the support values.

WHY ONE RULE IS ENOUGH

No figure in this program writes a literal color: they all draw through CSS properties, and their text through two or three classes. That was done so the figures would follow the light and the dark themes, and it has a second consequence that was worth exploiting: **redefining those properties restyles the whole application at once**. The studio writes **a single rule**, and no drawing module knows it exists. The rule is limited to `svg` elements on purpose: the colored labels of the interface use the same tokens, and a figure palette has no business recoloring the menus.

The **nine palettes** include the program's own, **Okabe–Ito** (color-blind safe), viridis, earth, cool, warm, pastels, high contrast and **grayscale**, which is there for a very specific question: if the journal charges for color, can your figure still be understood?

The **seven font families** are all on any ordinary computer: none is downloaded from the network, because a figure that falls back to another typeface on the reviewer's machine is a figure you did not design. The size goes from 0.7 to 1.6 times, the stroke from 0.5 to 3, and there are four backgrounds, among them transparent and white. One button puts everything back to the original.

> **WHAT IS EXPORTED IS WHAT YOU SEE**
>
> An SVG loaded as an image **has no style sheet**, so on export the colors have to be resolved by hand. The studio hooks in there: its colors win over the theme, its typeface replaces the one written in the code, and the chosen background is written as a rectangle beneath everything. It was checked by exporting a tree and **looking for the Okabe blue and the white background in the pixels of the PNG**.

12.7 The resolution the journal asks for

All the figures in the program are **SVG**, which has no resolution: it prints at any size without losing anything, and it is what you should send when the journal accepts it.

When it asks for a PNG at so many dots per inch, what it is really asking for is **a specific number of pixels**. The block does that arithmetic **before** generating anything and shows it: at 900 dpi a half-page figure is several thousand pixels on a side. And if the figure goes beyond what a browser can draw, **it says so and explains why**, instead of returning a blank image that looks like a bug in the program.

12.8 The methods, written with the numbers of the run

The methods section is the part of the paper that lets somebody else reach the same result, and it is the one written last, from memory, weeks later. Here the program writes it from what **actually ran**:

- A block that was not used **leaves no paragraph**.
- A support measure that was not computed **is not mentioned**.
- The number of replicates comes from the run, **not from a template**.
- The bibliography is built from what was cited and nothing else: the program knows 52 works and only the ones it used appear.

You choose the language, the title and the authorship, and whether the figures are included. The report is **a single self-contained HTML file**: no style sheet to fetch, no typeface to download, no script to run. It displays in any browser and prints to PDF from there.

AN HONEST DRAFT, NOT A FINISHED METHODS SECTION

What the program cannot write is **why** you chose what you chose: why that outgroup, why that fossil on that node, why that range cap. That is yours to add. The report saves you the parameters and the citations, which is where transcription errors are made, not the judgment.

12.9 The reproducible package

One file with everything: the data as they came in, the trees in the three formats, the figures in SVG with their colors already resolved, the tables in CSV, the report, and a text file that says **what each thing is and with what parameters it was made**. You choose which parts to include.

THE PROGRAM WRITES THE ZIP ITSELF

Byte by byte, with no external library and no network. That is why it works the same with the app opened by double-clicking, with no server and no connection—which is how you are going to use it the day you need it on someone else's computer.

12.10 How well it does, measured

This block does not compute: it **delivers**. So it was validated by having R do what whoever receives the package would do—open it and read it with their own tools. **38 checks** and not one disagreement.

- **The ZIP**: R's `unzip()` function lists and extracts every file, **checking each CRC32 along the way**, and the extracted sizes are the declared ones.
- **Newick and NEXUS**: read by `ape::read.tree` and `ape::read.nexus`, the same 18 tips, topological distance **0** between the two and branch lengths identical to 10^{-6}.
- **phyloXML**: parsed by `xml2`; its tip names are those of the tree and its branch lengths add up to what the Newick adds up to (2.446035 against 2.446036).
- **The alignment**: read by `ape::read.dna` as 18 × 1,711, with the taxa of the tree.
- **The report**: a file with no `<link>`, no `<script>` and no `<img>`, with its figures as inline SVG; and, in a package built from a distance tree alone, **citing no clock, no biogeography and no MCMC**, because none of them ran.
- **The figures**: well-formed XML with no unresolved `var(--…)`, which in Illustrator would be a black line.
- **The CRC32** against the value that the ZIP specification itself gives for the string "123456789": `0xCBF43926`.

WHAT TURNED UP DURING VALIDATION

The report module depended on the page's translator, which broke it outside the app and, worse, produced a **half-translated** document: the fragments it kept pasting in—"non-parametric bootstrap (100 replicates)," "by outgroup," "penalized likelihood"—were resolved **once**, in whatever language the interface was in, and then put into the Spanish paragraph **and** the English one. Now each fragment is stored as a pair, and the module no longer knows what language the page is in.

And one that is funny and instructive: the genome of a partition is stored as a two-letter code, which is reasonable in a menu and absurd in a methods section. It said "**The genomes represented are: nu**."

The figure studio produced two more. **The `.art-ink` class was used and had never been defined** in the style sheet—the tip names of the chronogram fell back to black, invisible in the dark theme—; and **the studio patched the export before the thing it was going to patch existed**, because its file was loaded *before* the one it wraps. The test page uncovered it, since it exports without waiting for the document to be ready.

12.11 Common mistakes

PUBLISHING A CLADOGRAM AND TALKING ABOUT IT AS A PHYLOGRAM

Without lengths, the figure says nothing about how much change there was. If the text talks about long branches or accelerated evolution, the figure has to show them.

RAISING THE SUPPORT THRESHOLD AND NOT SAYING SO

A tree from which the low support values have been erased looks much firmer than it is. Saying "support values ≥ 70 are shown" costs six words.

COLORING A PARAPHYLETIC GROUP WITHOUT WARNING

The panel tells you which ones are. A color in a figure reads as "lineage"; if it is not one, it has to be written in the caption.

ADDING A PHOTO WITHOUT CHECKING WHICH SPECIES IT IS

That is why there is a checkbox that cannot be skipped. The photo will be the first thing the reader looks at and the last thing they doubt.

SENDING A PNG WHEN THEY ACCEPT SVG

SVG has no resolution and prints at any size. A PNG at 300 dpi that is then enlarged in layout reaches the printer blurred.

HANDING IN THE REPORT AS IT IS

It brings the parameters and the right citations; it does not bring your reasons. Add why that outgroup, that fossil and that model before sending it.

KEEPING ONLY THE FIGURE

The package is small and contains the data as they came in, the trees in three formats and the index of what each thing is. It is what will save you when the reviewer asks, six months later, with what parameters it was made.

A

APPENDICES

Appendices

What you need to have at hand and did not fit in any chapter: which files the program reads and writes, the complete catalog of models with the citation of each matrix, all the decision rules of the twelve blocks gathered in a single table, the vocabulary, what to do when something does not work, the references and the license of the tool.

A The files the program reads and writes

B The catalog of models

C The decision rules, gathered

D Glossary

E When something goes wrong

F References

G The tool: license and how to cite it

A The files the program reads and writes

PhylogenyPro is a desktop program that runs in the browser and **sends nothing to any server**: every file comes from and goes to a folder on your computer. This appendix lists them.

A.1 · What it reads

FORMAT	WHERE AND WHAT FOR
FASTA `.fasta .fas .fa .fna`	Block 2. It is the format almost everybody starts with.
PHYLIP `.phy .phylip`	Block 2, in its four flavors: strict and relaxed, sequential and interleaved. The reader says which one it interpreted.
NEXUS `.nex .nexus`	Block 2. `DATA`, `CHARACTERS`, `SETS` and `TREES` blocks, with `MATCHCHAR` and comments.
Clustal `.aln` **MEGA** `.meg`	Block 2, alignments already made in another program.
GenBank `.gb .gbk .seq`	Block 2, the flat file just as it is downloaded.
Table `.csv .tsv .txt`	Block 2 for a morphological matrix; Block 9 for the characters, with one row per species and one column per character.
Partition scheme `.txt .part .nex .cfg`	Block 2. It reads the RAxML format and the NEXUS `SETS` block.
Trees `.tre .tree .newick .nwk .txt`	Block 11, one per line: those of each gene, a colleague's, a paper's.
Areas `.data .csv .tsv .phy .txt`	Block 10. The LAGRANGE format—the same one BioGeoBEARS uses—or a table with one column per area.
Pictures	Block 12, dragged into the manager. A file named after its taxon finds its own place.

> **A SPREADSHEET WORKBOOK IS NOT READ, AND ON PURPOSE**
>
> There is no reader for `.xlsx` or `.ods`. Adding one would mean putting into the program someone else's library of hundreds of thousands of lines, and it would stay inside forever. The choice was for the program to **load not a single line that is not its own**: that is exactly what lets it open with a double click, with nothing to install and no internet. Save the sheet as `.csv` and you are done.

A.2 · What it writes

FORMAT	WHAT COMES OUT AND FROM WHERE
FASTA, PHYLIP, NEXUS	Block 2: the alignment, trimmed or not. The NEXUS carries its `charsets`.
RAxML scheme	Block 2, to take the partitions to another program.
Newick	Blocks 4 to 12. Universal and narrow.
NEXUS of trees	Block 12, with a `TRANSLATE` block and the block that FigTree reads.

FORMAT	WHAT COMES OUT AND FROM WHERE
phyloXML	Block 12. The only one of the three that carries at once the support, the age and the interval of a node.
CSV	From almost every block: models, distances, support, concordance, ancestral ranges, events, splits, ABBA-BABA.
SVG	Every figure. No resolution, with the colors resolved and the pictures inside.
PNG	Every figure, at the dots per inch you ask for, with the pixel count shown before it is generated.
Report `.html`	Block 12. A single self-contained file; it prints to PDF from the browser.
Package `.zip`	Block 12. All of the above plus an index that says what each thing is.

A.3 · Keyboard shortcuts

The program is driven mostly with the mouse, but these shortcuts work in any block. `[` and `?` do not respond while you are typing in a field.

KEY	WHAT IT DOES
`Alt` + `←` / `→`	Previous or next block, from anywhere. In focused mode, previous or next section (at the ends, the block).
`←` / `→`, `Home`, `End`	With the focus on the block bar: move along it, or go to the first or the last block.
`Ctrl` + `K`	Opens Search: blocks, sections, controls and actions, even from a field.
`[`	Collapses or shows the block bar.
`?`	Shows the list of shortcuts and the help for where you are.
`Esc`	Closes dialogs and panels (interpretation help, image manager, list of shortcuts); while a long analysis is running, it works as Cancel.
`Enter` or space	On a card of the home page, opens that block.

B The catalog of models

B.1 · The 56 DNA models

Seven rate schemes, each with equal or empirical frequencies—fourteen base models—and each in the four variants of rate variation among sites: **plain, +I, +G and +I+G**. Fourteen times four, fifty-six.

EQUAL FREQ.	EMPIRICAL FREQ.	RATES	WHAT IT ADDS TO THE PREVIOUS ONE
JC	F81	0	The starting point: all changes equal.
K80	HKY	1	Transitions are told apart from transversions.
TrNef	TrN	2	The two transitions, with different rates.

EQUAL FREQ.	EMPIRICAL FREQ.	RATES	WHAT IT ADDS TO THE PREVIOUS ONE
K81	K81uf	2	Two classes of transversion.
TIMef	TIM	3	Different transitions and different transversions.
TVMef	TVM	4	The four transversions free.
SYM	GTR	5	All six rates free. There is nothing more general.

The **+I** adds a proportion of invariable sites; the **+G**, a discrete gamma with estimated α (Yang 1994), with the rates rescaled the way PAML does it. The right-hand column is, read from top to bottom, the chain of likelihood-ratio tests that the block can also run.

B.2 · The twelve protein matrices

Each with its 190 exchangeabilities and its 20 equilibrium frequencies, and each in the same variants, with **+F** to use the frequencies of your data instead of those of the matrix. All were verified against five independent canonical distributions.

MATRIX	WHAT IT WAS MADE FOR, AND ITS ORIGINAL PUBLICATION
LG	Nuclear proteins in general; the preferred one today. Le and Gascuel (2008).
WAG	Globular proteins. Whelan and Goldman (2001).
JTT	Dayhoff's successor, with much more data. Jones, Taylor and Thornton (1992).
Dayhoff	The first of them all, of historical value and still useful. Dayhoff, Schwartz and Orcutt (1978).
cpREV	**Chloroplast** proteins. Adachi, Waddell, Martin and Hasegawa (2000).
mtREV24	**Mitochondrial** proteins. Adachi and Hasegawa (1996).
VT	Derived from very divergent alignments. Müller and Vingron (2000).
RtREV	Retroviruses. Dimmic, Rest, Mindell and Goldstein (2002).
Blosum62	The classic scoring matrix, used here also as a model. Henikoff and Henikoff (1992).
FLU	Influenza viruses. Dang, Le, Gascuel and Le (2010).
HIVb	HIV, between hosts. Nickle and colleagues (2007).
mtMam	Mammalian mitochondria. Yang, Nielsen and Hasegawa (1998). Listed in the menu as `mtmam`.

B.3 · The other models, by block

WHERE	MODELS
Morphology Block 3	**Mk** and **Mkv** with Lewis's (2001) ascertainment correction, which is the one to use when the matrix has only variable characters.
Distances Block 3	p, JC69, K80, F81, TN93, LogDet (Lockhart et al. 1994) and paralinear (Lake), Poisson and Kimura for proteins, with their gamma-corrected variants.

WHERE	MODELS
Clock Block 7	Strict; penalized likelihood with λ by cross-validation (Sanderson 2002); least squares (To et al. 2016); and the Bayesian sampler's uncorrelated relaxed clock (Drummond et al. 2006).
Diversification Block 8	Pure birth, birth–death, two rates, density-dependent and exponential.
Characters Block 9	Continuous: Brownian, Ornstein–Uhlenbeck and early burst. Discrete: Mk in its **ER**, **SYM** and **ARD** variants.
Biogeography Block 10	**DEC** (Ree and Smith 2008), **DIVALIKE** and **BAYAREALIKE**, each with and without the founder event **+J** (Matzke 2013).

C The decision rules, gathered

The scales the program applies to each number, in a single place. On screen you do not need to look them up: the interpretation help paints the band next to the value. They are here so you can cite them, argue about them and take them to a meeting.

> **HALF OF THESE CUTOFFS ARE CUSTOMS, NOT LAWS**
>
> The 70% for the bootstrap, the 0.95 for the posterior, the 200 for the ESS: they are **reading conventions** that the community adopted, not thresholds derived from a theory. The program says so in every card where that is the case, and this appendix repeats it once for all. Use them to orient yourself, not to decide on their own.

C.1 · Data and alignment

QUANTITY	SCALE
Informative sites per taxon	<2 very few · 2–10 barely enough · 10–40 reasonable · >40 ample
Gaps in the matrix	<0.05 very clean · 0.05–0.2 normal · 0.2–0.5 much missing · >0.5 more gap than data
Trimming: columns removed	<0.1 almost nothing · 0.1–0.3 normal · 0.3–0.5 a lot · >0.5 the alignment was the problem
SP score	<0.7 badly aligned · 0.7–0.9 acceptable · 0.9–0.98 good · >0.98 almost perfect
Composition χ^2 (p)	<0.01 heterogeneous, careful · 0.01–0.05 doubtful · >0.05 compatible with homogeneity

C.2 · Models

QUANTITY	SCALE
ΔAIC from the best	0–2 just as good · 2–4 almost as good · 4–10 clearly worse · >10 ruled out
Akaike weight of the best	<0.5 no clear winner · 0.5–0.9 preferred · >0.9 outright winner

QUANTITY	SCALE
LRT (p)	<0.01 the complex one clearly wins · 0.01–0.05 the complex one wins · >0.05 the simple one is enough
α of the gamma	<0.5 very uneven · 0.5–1 uneven · 1–3 moderate · >3 almost homogeneous
κ (ts/tv)	<1.2 suspiciously low · 1.2–4 normal · 4–10 high · >10 very high

C.3 · Trees and support

QUANTITY	SCALE
CI (consistency)	<0.3 much homoplasy · 0.3–0.5 considerable · 0.5–0.75 moderate · >0.75 little
RI (retention)	<0.4 little structure · 0.4–0.6 some · 0.6–0.8 good · >0.8 very good
Bootstrap (%)	<50 unsupported · 50–70 weak · 70–95 supported · >95 strong
UFBoot (%)	<85 unsupported · 85–95 doubtful · >95 supported
SH-aLRT (%)	<80 unsupported · >80 supported
AU test (p)	<0.05 rejected · 0.05–0.1 borderline · >0.1 not rejected
Normalized RF	<0.1 almost the same tree · 0.1–0.3 minor differences · 0.3–0.6 they disagree · >0.6 different topologies

> **THE THREE SUPPORT SCALES ARE NOT THE SAME**
>
> A **70% bootstrap**, a **95% UFBoot** and a **posterior of 0.95** say roughly the same thing. Reading UFBoot with the bootstrap's yardstick—celebrating a 75—is the most common mistake with the ultrafast bootstrap. They are in three separate rows for that reason.

C.4 · Bayesian

QUANTITY	SCALE
Posterior	<0.5 unsupported · 0.5–0.95 doubtful · >0.95 supported
ESS	<100 insufficient · 100–200 barely enough · >200 sufficient
PSRF ($\hat{R}$)	<1.01 converged · 1.01–1.05 borderline · >1.05 did not converge
ASDSF	<0.01 converged · 0.01–0.05 not yet · >0.05 far off
2·ln(BF)	<2 negligible · 2–6 positive · 6–10 strong · >10 very strong (Kass and Raftery 1995)

C.5 · Time, diversification and characters

QUANTITY	SCALE
Root-to-tip **r²**	<0.2 no temporal signal · 0.2–0.5 weak · 0.5–0.8 clear · >0.8 very marked clock
Strict clock (p)	<0.05 rejected · >0.05 not rejected
Rate variation (CV)	<0.1 practically a clock · 0.1–0.3 moderate · 0.3–0.6 high · >0.6 very high
Pybus and Harvey's **γ**	<–1.645 slowdown · –1.645 to 1.645 compatible with a constant rate · >1.645 speedup
ε = μ/λ	<0.2 little extinction · 0.2–0.6 moderate · 0.6–0.9 high · >0.9 almost everything goes extinct
Pagel's **λ**	<0.2 no signal · 0.2–0.6 partial · 0.6–0.9 good · >0.9 Brownian
Blomberg's **K**	<0.5 less signal than Brownian · 0.5–1 somewhat less · >1 more than Brownian

C.6 · Discordance

QUANTITY	SCALE
gCF (%)	<33 most disagree · 33–50 strong conflict · 50–75 majority in favor · >75 broad agreement
sCF (%)	<36 no signal · 36–45 very spread out · 45–60 majority · >60 clear
Jackknife **|z|** (D)	<2 no evidence · 2–3 suggestive · >3 gene flow

D Glossary

Alignment (*alineamiento*)

The hypothesis that the columns of a sequence matrix are homologous positions. It is not a neutral preliminary step: it is the first result of the study.

Ancestral state (*estado ancestral*)

The reconstructed value of a character at an internal node. It is reported as probabilities, not as a winner.

ASDSF

Average standard deviation of split frequencies between independent runs. It measures whether two chains arrived at the same place.

Bayes factor (*factor de Bayes*)

The ratio of the marginal likelihoods of two models. It compares by integrating over all the values of the parameters, not with a shortcut like the AIC.

Block jackknife (*jackknife por bloques*)

Error estimated by removing contiguous blocks of sites, because neighboring sites are not independent.

Blomberg's K

Compares the observed variance of a character with the one Brownian motion predicts on that tree. It measures magnitude, not shape.

Bootstrap (*arranque*)

Resampling columns with replacement. How often a branch reappears is its bootstrap support.

Branch length (*longitud de rama*)

The expected amount of change per site along that branch. In a chronogram, time.

Brownian motion (*movimiento browniano*)

A model of pure drift for a continuous character: the variance between two species grows with the time they have been apart.

Calibration justification (*ensayo de calibración*)

Putting in writing which fossil, which event or which rate each age comes from, with its prior. It is the most debated part of a dating study.

Chronogram (*cronograma*)

A tree whose axis is time, not change. It requires a clock and at least one calibration.

Clade (*clado*)

An ancestor and all its descendants. A group that leaves any of them out is paraphyletic.

Cladogenesis (*cladogénesis*)

What happens at the node when a lineage splits. In biogeography it is where the range is divided between the daughters.

Cladogram (*cladograma*)

A tree drawn without branch lengths: it asserts relationship and nothing more.

Concordance (gCF, sCF)

What proportion of the gene trees, or of the sites, contains a branch. It is a question different from support.

Consensus (*consenso*)

A summary tree of a set of trees. Strict, majority-rule, semi-strict or greedy; all of them throw away what contradicts itself.

Convergence (of a chain)

The sampler is already drawing samples from the posterior distribution and not from the path toward it. It is diagnosed with ESS, PSRF and ASDSF, never with just one.

Convergence (phylogenetic)

Two lineages that reach the same state by separate routes. It is the source of homoplasy.

DEC

The dispersal–extinction–cladogenesis model for geographic ranges (Ree and Smith 2008).

Discrete gamma (*gamma discreta*)
Rate variation among sites divided into a few categories, with a single shape parameter α (Yang 1994).

Early burst (*estallido temprano*)
A model in which the rate of evolution of a character decays with time; the model of adaptive radiation.

ESS
Effective sample size: how many *independent* samples your chain is worth.

Founder event (+J) (*salto fundador*)
A daughter that, at the split, leaves for a single area outside the ancestor's range (Matzke 2013).

Homoplasy (*homoplasia*)
Similarity that does not come from a common ancestor: convergence, parallelism or reversal.

Independent contrasts (*contrastes independientes*)
Differences between sister nodes, scaled by their branches, that turn n non-independent species into n–1 observations that are independent (Felsenstein 1985).

Introgression (*introgresión*)
The passage of genetic material between lineages through hybridization. It leaves an asymmetry that incomplete lineage sorting does not leave.

Maximum likelihood (*máxima verosimilitud*)
The criterion that picks the tree and parameters that make the observed data most probable under the model.

MCMC
Markov chain Monte Carlo: the algorithm with which the Bayesian posterior is sampled.

Mk
A k-state model for discrete characters (Lewis 2001); Mkv adds the correction for having sampled only variable characters.

Molecular clock (*reloj molecular*)
The assumption that change accumulates at a comparable rate across lineages. Strict if it is a single rate; relaxed if it varies from branch to branch.

Monophyletic (*monofilético*)
Said of a group that is a complete clade.

Null range (*rango nulo*)
The state of a lineage that has lost its last area. It is absorbing, and it is what makes extinction real in the model.

Ornstein–Uhlenbeck
A model with a force that pulls the character toward an optimum. Its α is read as a half-life against the depth of the tree.

Outgroup (*exogrupo*)

A taxon known to be outside the study group, used to place the root. A bad choice shifts the whole history.

Pagel's λ

Scales the covariances between species without touching the variances. It measures shape, not magnitude.

Paraphyletic (*parafilético*)

A group that excludes some of the descendants of its common ancestor.

Parsimony (*parsimonia*)

The criterion that prefers the tree that requires the fewest changes.

Partition (*partición*)

A subset of columns with its own model: a gene, a codon position, a genome.

PGLS

Generalized least-squares regression whose covariance matrix is that of the tree. It gives the same slope as the contrasts under Brownian motion.

Phylogenetic signal (*señal filogenética*)

How much more relatives resemble each other than expected by chance. A pattern, not a process.

Phylogram (*filograma*)

A tree with branch lengths: it also asserts how much change there was.

Polytomy (*politomía*)

A node with more than two daughters. It almost always means "I don't know in what order," not "they split at the same time."

Posterior

The probability of a branch given the data, the model and the prior. It is not on the same scale as a bootstrap support.

Prior

What is asserted before seeing the data. In dating, the shape that declares the uncertainty of an age.

PSRF

Potential scale reduction factor ($\hat{R}$): it compares the variation within each chain with the variation between chains.

Pybus and Harvey's γ

Compares where the nodes of a tree fall with where they would fall under a constant rate. Negative, slowdown; but missing species mimic it exactly.

Quartet (*cuarteto*)

A group of four tips. It has three possible resolutions, and counting them is the basis of the species tree and of a distance between trees.

Robinson–Foulds distance
: The number of splits present in one tree and absent from the other. It breaks easily: one misplaced tip takes it to its maximum.

Saturation (*saturación*)
: The state in which so many changes have piled up that the observed difference no longer grows with time.

Split (*división*)
: The division of the tips into two sets produced by an internal branch. An unrooted tree is its set of splits.

Split network (*red de divisiones*)
: A figure that keeps the incompatible splits instead of throwing them away, drawing them as boxes.

Stochastic mapping (*mapeo estocástico*)
: Drawing complete histories of the character on the tree, conditioned on the tips. It turns probabilities into counts of events.

Support (*apoyo*)
: A measure of how confidently *these* data hold up a branch. It does not say that the branch is true, nor how many genes see it.

Tanglegram (*tanglegrama*)
: Two trees facing each other, joined tip to tip. No crossings does not mean they are equal.

Ultrametric (*ultramétrico*)
: A tree in which all the tips lie at the same distance from the root. A chronogram of living species is one.

Vicariance (*vicarianza*)
: The division of a range in two by a barrier. In DEC one of the daughters keeps a single area; in DIVALIKE the cut can be of any size.

E When something goes wrong

WHAT YOU SEE	WHAT IT USUALLY IS
"No row in the table" (Block 9) or **"No areas in the table for"** (Block 10), followed by tip names	Those tips of the tree were not found in the table. Both blocks already forgive spaces against underscores (`Bursera_palmeri` against `Bursera palmeri`) and capitals, so it is almost always a typo, an abbreviation or a missing row. Fix the file, not the list. Block 9 prunes the tree to the tips that have the character; Block 10 needs a range for every tip.
The **tree comes out with a huge polytomy**	A consensus with many splits below the threshold. It is not an error: the data do not resolve that node.
The **Bayesian chain does not converge**	Look at the ESS and the PSRF before anything else. If the ESS is low in only one parameter, running longer is usually enough; if the PSRF is high, the chains are in different places and the problem is the model or the prior.

WHAT YOU SEE	WHAT IT USUALLY IS
The **branches come out extremely long** in the Bayesian analysis	The branch-length prior. With divergent data, an exponential with mean 0.1 drags the whole tree down. The panel compares with the maximum-likelihood tree before running: heed it.
The **OU wins** but the half-life is enormous	Indistinguishable from Brownian motion. Do not conclude stabilizing selection.
A **+J model wins** by a lot	It is to be expected. Report by how much it wins and over which model; do not turn it into long-distance dispersal without more.
The **PNG comes out blank** or the program refuses	You asked for more pixels than a browser can draw. Lower the dots per inch, or send the SVG, which has no resolution.
The **pictures do not appear** on the tree	The confirmation checkbox of each one has not been ticked. It is deliberate.
The **colors of the exported SVG** are not those on screen	It should not happen, and there are tests watching for it; if it does, export again from Block 12, which is where the figure studio hooks into the export.
The **report cites something you did not run**	It should not: the report is built from what actually ran. If you see it, it is a bug and worth reporting.
A **block takes forever**	Check the number of taxa and sites against the table of sizes on the home page. Heavy computations run in separate threads, so the page does not freeze even if it seems not to be moving.
You lost everything on closing	The program does not save on its own. Build the Block 12 package before closing; it is a single button.

F References

The works cited in this manual. The program knows many more and builds the bibliography of the report from the ones your study actually used.

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G The tool: license and how to cite it

G.1 · What it is and what it is not

PhylogenyPro is a program **written entirely from scratch**: it loads no third-party library, no file from a content delivery network, no minified code. Everything it does—the likelihood engine, the sampler, the drawings, the ZIP writer—is in its own code and can be read.

What it does contain from outside are **published numerical constants**: the twelve amino-acid matrices, the aligner's BLOSUM62 matrix, the genetic code and the geological time scale. They are not anyone's creative expression: they are the parameters of published scientific models, and the file `PROCEDENCIA.md` documents where each one comes from and how it was verified.

G.2 · License

It is distributed under the **GNU General Public License, version 3 or later** (GPL-3.0-or-later). The full text is in the file `LICENSE`, at the root of the program.

> **I LICENSING IS NOT GIVING UP AUTHORSHIP**
>
> Choosing a free license is a decision about **what whoever receives the program may do**. Authorship is neither transferred nor waived: the author's moral rights are inalienable. You may use the program, study it, modify it and redistribute it; what the GPL asks in return is that, if you distribute a modified version, you distribute it with the same freedoms.

G.3 · How to cite it

At the root of the program there are a `CITATION.cff` file and a `codemeta.json` file with the complete, machine-readable details; many reference managers read them directly. The short form, for the program and for this manual:

> **∑ SUGGESTED CITATION**
>
> Barrera-Guzmán, L.Á. (2026). *PhylogenyPro: molecular phylogenetics in the browser, from the alignment to the dated tree* (version 1.4.0). Universidad Autónoma Chapingo. Zenodo. https://doi.org/10.5281/zenodo.23005815
>
> Barrera-Guzmán, L.Á. (2026). *PhylogenyPro: User's Manual* (1st English ed.). Distributed with PhylogenyPro, version 1.4.0. https://doi.org/10.5281/zenodo.23005815
>
> That is the **concept DOI**: it always leads to the most recent version, and it is the one to cite. To pin down the exact version you ran—the right thing when the result has to be repeatable—take its own DOI from the Zenodo record; for 1.4.0 it is `10.5281/zenodo.23005816`.

In the methods section, also cite **the works behind the methods you used**: the substitution model, the type of support, the dating method, the biogeographic model. The Block 12 report gathers them on its own, and only the ones your study actually used, which is exactly the list that has to go there.